



**ŘEDITELSTVÍ SILNIC A DÁLNIC ČR**

# **NDIC - DATEX II Situation Publication - Restrictions**

*Release 1.1rev0*

**Národní Dopravní Informační Centrum**

**2018-04-29**

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The specification defines format for publishing information about planned and actual closures and traffic restrictions.

## INTRODUCTION

This document shall be used by publisher and / or subscriber. Subscriber shall use it for format suitability evaluation and for possible implementation of received NDIC data import into own information system. Publisher shall use it as a specification of data to be published format and for possible implementation of data export from own information system into NDIC.

Here you can find:

- conceptual description of published information
- information about location referencing methods used
- W3C XML schema of the format
- data sample(s)

Following topics are out of scope of this document and if present, shall be considered as informative only:

- transfer protocol incl. access point, life cycle etc.
- related data formats
- typical data origin
- typical data use

### 1.1 General terms

Following terms are used in this document:

**format**

description of data structure for transferring some kind of information

**protocol**

description of data transfer process and rules

**NDIC**

national traffic information centre

**DATEX II model**

class model for describing information related to the road traffic. The model in form of UML is platform independent.

One of specific platforms for expressing the model is XML.

The model is extensive (it consists of few thousands of structural items representing term incl. definition) For reference data (UML class model in form of Enterprise Architect EAP file, XML schema etc.) see [\[d2ref\]](#). For additional information see [\[d2supp\]](#).

**(DATEX II) profile**

simplified class model, still sufficient for specific needs. Profile is created from model by selecting only specific parts.

**(DATEX II) extension**

extended DATEX II model, adding data structures missing in standard model but needed for specific needs.

**B level extension**

Extension of DATEX II model, which is backward compatible with the original DATEX II model, even though it contains additional structures. [16157-1]

**C level extension**

Extension of DATEX II model, which is not backward compatible with the original DATEX II model.

**publication**

an element of DATEX II model, serving as an envelope for transmitted content of certain kind. Every sent DATEX II message contains exactly one element of publication type. There are various types of publications, for example *SituationPublication* [16157-3], *ElaboratedDataPublication* [16157-5], *PredefinedLocationsPublication* [16157-3] etc.

**(W3C) XML schema**

language, which allows definition of rules for a XML document structure.

**XSD**

XML document containing XML schema. Sometimes it is synonym for XML schema itself.

**uri**

uniform resource identifier, text, used as unique identifier. In this document, every format has it's own uri.

## CONCEPTS

Described format allows publishing information about road planned closures and restrictions.

### 2.1 Publication type SituationPublication

Format is based on DATEX II publication of type *SituationPublication* [16157-3]. Profile uses only small subset of possible structures.

```
1 <?xml version="1.0" encoding="utf-8"?>
2 <d2LogicalModel modelBaseVersion="2"
3 xmlns="http://datex2.eu/schema/2/2_0"
4 xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
5 xsi:schemaLocation="http://datex2.eu/schema/2/2_0 ../schema/schema.xsd">
6
7   <exchange>
8     <supplierIdentification>
9       <country>cz</country>
10      <nationalIdentifier>NDIC</nationalIdentifier>
11    </supplierIdentification>
12  </exchange>
13  <payloadPublication xsi:type="SituationPublication" lang="cz">
14    <feedType>x-format:cz-ndic_d2-restrictions-v1.1</feedType>
15    <publicationTime>2017-07-01T23:11:30+02:00</publicationTime>
16    <publicationCreator>
17      <country>cz</country>
18      <nationalIdentifier>NDIC</nationalIdentifier>
19    </publicationCreator>
```

### 2.2 Event description (only OperatorActions)

*SituationPublication* requires description by means of class *Situation*, which aggregates instances of class *SituationRecord*, namely descendants of class *OperatorActions*. *Situation* also contains elements with general service availability information.

Element *situation* describes one particular event, more particular events means more *situation* elements. Since the situation described may evolve over time, the element is versioned and has unique identifier. Each *situation* element contains one or more *situationRecord* elements.

The *situationRecord* element may, in the element *situation*, occur several times (each element describing different aspect of the situation, i.e. accident in one record leading to a queue described in another record). The *situationRecord* element may evolve over time, so the element is versioned and has unique identifier.

```

20 <situation id="ff808181-49e6-e823-014a-07e78fe011e5" version="2">
21   <headerInformation>
22     <confidentiality>noRestriction</confidentiality>
23     <informationStatus>real</informationStatus>
24     <urgency>normalUrgency</urgency>
25   </headerInformation>
26   <situationRecord xsi:type="RoadOrCarriagewayOrLaneManagement"
27     id="ff808181-49e6-e823-014a-07e78fe011e5" version="1">
28     <situationRecordCreationTime>2017-07-01T23:05:00+02:00</
↪ situationRecordCreationTime>
29     <situationRecordVersionTime>2017-07-01T23:11:30+02:00</
↪ situationRecordVersionTime>
30     <probabilityOfOccurrence>certain</probabilityOfOccurrence>
31     <source>
32       <sourceIdentification>NDIC</sourceIdentification>
33       <sourceName>
34         <values>
35           <value lang="cs">Operátoři NDIC</value>
36         </values>
37       </sourceName>
38     </source>
39     <validity>
40       <validityStatus>definedByValidityTimeSpec</validityStatus>
41       <validityTimeSpecification>
42         <overallStartTime>2017-07-01T23:05:00+02:00</overallStartTime>
43         <overallEndTime>2017-07-02T10:00:00+02:00</overallEndTime>
44       </validityTimeSpecification>
45     </validity>
46     <impact>
47       <capacityRemaining>0</capacityRemaining>
48     </impact>
49     <generalPublicComment>
50       <comment>
51         <values>
52           <value lang="cs">Uzavírka komunikace pro vozidla nad 3.5
53             t</value>
54         </values>
55       </comment>
56     </generalPublicComment>
57     <groupOfLocations xsi:type="Linear">
58       <alertCLinear xsi:type="AlertCMethod2Linear">
59         <alertCLocationCountryCode>CZ</alertCLocationCountryCode>
60         <alertCLocationTableNumber>25</alertCLocationTableNumber>
61         <alertCLocationTableVersion>6.0</alertCLocationTableVersion>
62         <alertCDirection>
63           <alertCDirectionCoded>negative</alertCDirectionCoded>
64         </alertCDirection>
65         <alertCMethod2PrimaryPointLocation>
66           <alertCLocation>
67             <specificLocation>25532</specificLocation>
68           </alertCLocation>
69         </alertCMethod2PrimaryPointLocation>
70         <alertCMethod2SecondaryPointLocation>
71           <alertCLocation>
72             <specificLocation>25531</specificLocation>
73           </alertCLocation>
74         </alertCMethod2SecondaryPointLocation>

```

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```

75     </alertCLinear>
76     <linearWithinLinearElement>
77         <directionRelativeOnLinearSection>opposite</
↪ directionRelativeOnLinearSection>
78         <linearElement>
79             <roadNumber>D1</roadNumber>
80         </linearElement>
81         <fromPoint xsi:type="DistanceFromLinearElementStart">
82             <distanceAlong>18500</distanceAlong>
83         </fromPoint>
84         <toPoint xsi:type="DistanceFromLinearElementStart">
85             <distanceAlong>17000</distanceAlong>
86         </toPoint>
87     </linearWithinLinearElement>
88     <globalNetworkLinear>
89         <networkVersion>
90             <values>
91                 <value lang="cs">15.06</value>
92             </values>
93         </networkVersion>
94         <linearGeometryType>continuous</linearGeometryType>
95         <startPoint>
96             <sjtskPointCoordinates>
97                 <sjtskX>-726024</sjtskX>
98                 <sjtskY>-1062544</sjtskY>
99             </sjtskPointCoordinates>
100        </startPoint>
101        <endPoint>
102            <sjtskPointCoordinates>
103                <sjtskX>-727210</sjtskX>
104                <sjtskY>-1061636</sjtskY>
105            </sjtskPointCoordinates>
106        </endPoint>
107        <linearWithinLinearGNElement>
108            <sectionId>1687624</sectionId>
109            <directionRelativeOnLinearSection>opposite</
↪ directionRelativeOnLinearSection>
110            <orderOfSection>0</orderOfSection>
111            <fromPoint xsi:type="PercentageDistanceAlongLinearElement">
112                <percentageDistanceAlong>
113                    0.349030470911379</percentageDistanceAlong>
114            </fromPoint>
115            <toPoint xsi:type="PercentageDistanceAlongLinearElement">
116                <percentageDistanceAlong>1</percentageDistanceAlong>
117            </toPoint>
118        </linearWithinLinearGNElement>
119        <linearWithinLinearGNElement>
120            <sectionId>1601936</sectionId>
121            <directionRelativeOnLinearSection>opposite</
↪ directionRelativeOnLinearSection>
122            <orderOfSection>1</orderOfSection>
123            <fromPoint xsi:type="PercentageDistanceAlongLinearElement">
124                <percentageDistanceAlong>0</percentageDistanceAlong>
125            </fromPoint>
126            <toPoint xsi:type="PercentageDistanceAlongLinearElement">
127                <percentageDistanceAlong>1</percentageDistanceAlong>

```

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```

128         </toPoint>
129     </linearWithinLinearGNElement>
130     <linearWithinLinearGNElement>
131         <sectionId>3068685</sectionId>
132         <directionRelativeOnLinearSection>opposite</
↪ directionRelativeOnLinearSection>
133         <orderOfSection>2</orderOfSection>
134         <fromPoint xsi:type="PercentageDistanceAlongLinearElement">
135             <percentageDistanceAlong>0</percentageDistanceAlong>
136         </fromPoint>
137         <toPoint xsi:type="PercentageDistanceAlongLinearElement">
138             <percentageDistanceAlong>1</percentageDistanceAlong>
139         </toPoint>
140     </linearWithinLinearGNElement>
141     <linearWithinLinearGNElement>
142         <sectionId>3068686</sectionId>
143         <directionRelativeOnLinearSection>opposite</
↪ directionRelativeOnLinearSection>
144         <orderOfSection>3</orderOfSection>
145         <fromPoint xsi:type="PercentageDistanceAlongLinearElement">
146             <percentageDistanceAlong>0</percentageDistanceAlong>
147         </fromPoint>
148         <toPoint xsi:type="PercentageDistanceAlongLinearElement">
149             <percentageDistanceAlong>
150                 0.00793650792934262</percentageDistanceAlong>
151         </toPoint>
152     </linearWithinLinearGNElement>
153 </globalNetworkLinear>
154 </groupOfLocations>
155 <complianceOption>mandatory</complianceOption>
156 <forVehiclesWithCharacteristicsOf>
157     <vehicleType>lorry</vehicleType>
158     <grossWeightCharacteristic>
159         <comparisonOperator>greaterThan</comparisonOperator>
160         <grossVehicleWeight>3.5</grossVehicleWeight>
161     </grossWeightCharacteristic>
162 </forVehiclesWithCharacteristicsOf>
163 <roadOrCarriagewayOrLaneManagementType>roadClosed</
↪ roadOrCarriagewayOrLaneManagementType>
164 </situationRecord>
165 </situation>

```

## 2.3 Location description (Point or linear road network segment)

*Location* is a point or linear segment, both referenced to a road network segment. Location of type area and single point locations without a road segment reference are not used. The location may be described by means of several methods (at least by one). Each *SituationRecord* has to have its own location description. Within one *Situation*, the *SituationRecord* location descriptions, may differ in i.e. one *SituationRecord* describing point source that caused a linear problem (second *SituationRecord*).

```

57 <groupOfLocations xsi:type="Linear">
58     <alertCLinear xsi:type="AlertCMethod2Linear">
59         <alertCLocationCountryCode>CZ</alertCLocationCountryCode>
60         <alertCLocationTableNumber>25</alertCLocationTableNumber>

```

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```

61      <alertCLocationTableVersion>6.0</alertCLocationTableVersion>
62      <alertCDirection>
63        <alertCDirectionCoded>negative</alertCDirectionCoded>
64      </alertCDirection>
65      <alertCMethod2PrimaryPointLocation>
66        <alertCLocation>
67          <specificLocation>25532</specificLocation>
68        </alertCLocation>
69      </alertCMethod2PrimaryPointLocation>
70      <alertCMethod2SecondaryPointLocation>
71        <alertCLocation>
72          <specificLocation>25531</specificLocation>
73        </alertCLocation>
74      </alertCMethod2SecondaryPointLocation>
75    </alertCLinear>
76    <linearWithinLinearElement>
77      <directionRelativeOnLinearSection>opposite</
↪ directionRelativeOnLinearSection>
78      <linearElement>
79        <roadNumber>D1</roadNumber>
80      </linearElement>
81      <fromPoint xsi:type="DistanceFromLinearElementStart">
82        <distanceAlong>18500</distanceAlong>
83      </fromPoint>
84      <toPoint xsi:type="DistanceFromLinearElementStart">
85        <distanceAlong>17000</distanceAlong>
86      </toPoint>
87    </linearWithinLinearElement>
88    <globalNetworkLinear>
89      <networkVersion>
90        <values>
91          <value lang="cs">15.06</value>
92        </values>
93      </networkVersion>
94      <linearGeometryType>continuous</linearGeometryType>
95      <startPoint>
96        <sjtskPointCoordinates>
97          <sjtskX>-726024</sjtskX>
98          <sjtskY>-1062544</sjtskY>
99        </sjtskPointCoordinates>
100      </startPoint>
101      <endPoint>
102        <sjtskPointCoordinates>
103          <sjtskX>-727210</sjtskX>
104          <sjtskY>-1061636</sjtskY>
105        </sjtskPointCoordinates>
106      </endPoint>
107      <linearWithinLinearGNElement>
108        <sectionId>1687624</sectionId>
109        <directionRelativeOnLinearSection>opposite</
↪ directionRelativeOnLinearSection>
110        <orderOfSection>0</orderOfSection>
111        <fromPoint xsi:type="PercentageDistanceAlongLinearElement">
112          <percentageDistanceAlong>
113            0.349030470911379</percentageDistanceAlong>
114          </fromPoint>

```

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```

115         <toPoint xsi:type="PercentageDistanceAlongLinearElement">
116             <percentageDistanceAlong>1</percentageDistanceAlong>
117         </toPoint>
118     </linearWithinLinearGNElement>
119     <linearWithinLinearGNElement>
120         <sectionId>1601936</sectionId>
121         <directionRelativeOnLinearSection>opposite</
122     <directionRelativeOnLinearSection>
123         <orderOfSection>1</orderOfSection>
124         <fromPoint xsi:type="PercentageDistanceAlongLinearElement">
125             <percentageDistanceAlong>0</percentageDistanceAlong>
126         </fromPoint>
127         <toPoint xsi:type="PercentageDistanceAlongLinearElement">
128             <percentageDistanceAlong>1</percentageDistanceAlong>
129         </toPoint>
130     </linearWithinLinearGNElement>
131     <linearWithinLinearGNElement>
132         <sectionId>3068685</sectionId>
133         <directionRelativeOnLinearSection>opposite</
134     <directionRelativeOnLinearSection>
135         <orderOfSection>2</orderOfSection>
136         <fromPoint xsi:type="PercentageDistanceAlongLinearElement">
137             <percentageDistanceAlong>0</percentageDistanceAlong>
138         </fromPoint>
139         <toPoint xsi:type="PercentageDistanceAlongLinearElement">
140             <percentageDistanceAlong>1</percentageDistanceAlong>
141         </toPoint>
142     </linearWithinLinearGNElement>
143     <linearWithinLinearGNElement>
144         <sectionId>3068686</sectionId>
145         <directionRelativeOnLinearSection>opposite</
146     <directionRelativeOnLinearSection>
147         <orderOfSection>3</orderOfSection>
148         <fromPoint xsi:type="PercentageDistanceAlongLinearElement">
149             <percentageDistanceAlong>0</percentageDistanceAlong>
150         </fromPoint>
151         <toPoint xsi:type="PercentageDistanceAlongLinearElement">
152             <percentageDistanceAlong>
153                 0.00793650792934262</percentageDistanceAlong>
154         </toPoint>
155     </linearWithinLinearGNElement>
156 </globalNetworkLinear>
157 </groupOfLocations>

```

## 2.4 Time validity

Time validity is stated independently for every single event (*SituationRecord*) described in a situation and is defined by a time interval between two time instants. The class *Validity* uses exclusively an instance of class *OverallPeriod* in the role of *validityTimeSpecification* and elements *overallStartTime* and *overallStopTime* to define start and end of the interval.

Setting a week-day and recurring time validity within the specified period, e.g. for recurring intermittent nightly tunnel closure, is not allowed. For such information, splitting to more messages valid in respective times is used.

Used time validity intervals may fall into past, present or future.

```

39      <validity>
40        <validityStatus>definedByValidityTimeSpec</validityStatus>
41        <validityTimeSpecification>
42          <overallStartTime>2017-07-01T23:05:00+02:00</overallStartTime>
43          <overallEndTime>2017-07-02T10:00:00+02:00</overallEndTime>
44        </validityTimeSpecification>
45      </validity>

```

## 2.5 Location description

The locations of the events varies thus its description can not be prepared in advance. For this reason, description by reference to *PrefefinedLocationPublication* can not be used. Element *groupOfLocations* with type *Point* or *Linear* (descendants of *NetworkLocation*) contains description of one physical location in one or more location referencing methods.

```

57      <groupOfLocations xsi:type="Linear">

```

Three location description methods for a linear location and four location description methods for point location can be used to describe the location. The method use depends on method availability at the location. At least one location description should be used. Point description without a linear reference is not allowed.

Following is a selection of most used location referencing methods.

### 2.5.1 Location description by means of ALERT-C

The method describes position by ALERT-C codes, by class *alertCLinear* of the *AlertCMethod2Linear* type for linear locations and by class *alertCPoint* of the *AlertCMethod2Point* type for point locations. Other methods of ALERT-C description are not used. The use of this method depends on the presence of the ALERT-C location at a given location.

The *alertCLinear* method uses the primary location (element *alertCMethod2PrimaryPointLocation*), secondary location (element *alertCMethod2SecondaryPointLocation*) and direction (*alertCDirection*) element to describe the event. Of course, the method also includes elements that allows to identify a particular location table.

Element *alertCLinear* can appear more than once in a situation location description. This could happen for long events that must be described by more than one ALERT-C route. ALERT-C allows to chain (reference between) locations only within one route.

The primary location is, by the ALERT-C principle, the place where the accident occurred or where the driver leaves the problem area. The secondary location is the end of the traffic queue or the place where the driver first enters the problem area.

```

58      <alertCLinear xsi:type="AlertCMethod2Linear">
59        <alertCLocationCountryCode>CZ</alertCLocationCountryCode>
60        <alertCLocationTableNumber>25</alertCLocationTableNumber>
61        <alertCLocationTableVersion>6.0</alertCLocationTableVersion>
62        <alertCDirection>
63          <alertCDirectionCoded>negative</alertCDirectionCoded>
64        </alertCDirection>
65        <alertCMethod2PrimaryPointLocation>
66          <alertCLocation>
67            <specificLocation>25532</specificLocation>
68          </alertCLocation>
69        </alertCMethod2PrimaryPointLocation>
70        <alertCMethod2SecondaryPointLocation>
71          <alertCLocation>

```

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```

72      <specificLocation>25531</specificLocation>
73      </alertCLocation>
74      </alertCMethod2SecondaryPointLocation>
75      </alertCLinear>

```

This method is not optimal because the RDS-TMC uses to describe the event location the primary location, direction and extent, everything else can be found in the location table. Therefore, if the recipient systems are adapted to RDS-TMC event reconstruction, it will be necessary to modify them.

The *alertCPoint* method is used to describe the point location (the inside of a junction or a tunnel, a short section). This method uses the primary location (element *alertCMethod2PrimaryPointLocation*) and the direction (*alertCDirection*) to describe the event. Of course, the method also includes elements that allow to identify a particular location table.

However, if it is necessary to express the short location covered by one ALERT-C point also by other methods, then the *alertCLinear* method has to be used (where the secondary location is equal to the primary), since the methods describing one location must be of same type e.g. point or linear.

## 2.5.2 Location description using a textual description of the road network

The method uses the “address” of the street or road (name, area, number, etc.) and the distance of the event location from the beginning and the end of the road. For description of linear locations, the *linearWithinLinearElement* class is used and for point locations the *pointAlongLinearElement* class is used. The use of this method depends on the presence of a proper description of the road (ideal for motorways).

The *roadNumber* element describes the number of the affected road, element *DirectionRelativeOnLinearSection* describes the direction of affected traffic, and the *fromPoint* and *toPoint* elements describes distance of the start and the end of the event from the beginning of the road.

```

76      <linearWithinLinearElement>
77          <directionRelativeOnLinearSection>opposite</
78      ↪directionRelativeOnLinearSection>
79          <linearElement>
80              <roadNumber>D1</roadNumber>
81          </linearElement>
82          <fromPoint xsi:type="DistanceFromLinearElementStart">
83              <distanceAlong>18500</distanceAlong>
84          </fromPoint>
85          <toPoint xsi:type="DistanceFromLinearElementStart">
86              <distanceAlong>17000</distanceAlong>
87          </toPoint>
      </linearWithinLinearElement>

```

The method for description of a point event *PointAlongLinearElement* differs by simply describing one distance from the beginning of the identified road instead of two.

## 2.5.3 Location description using the Global Network extension

This is a DATEX II model extension type B.

The method uses for location description identifiers of affected sections of the road network, specified by Global Network (GN) database, and event start and end coordinates (in the S-JTSK reference system). The last and initial segments are truncated by the relative distance of the start and end points of the event relative to the GN segment.

The *globalNetworkLinear* class is used to describe the linear location and *pointWithinLinearGNElement* for point location description.

The *globalNetworkLinear* element describes the location of one event on the network. It contains the network version identifier *networkVersion*, the points bounding the beginning and end of the event (*startPoint* and *endPoint* in coordinates, here S-JTSK). And one or more *linearWithinLinearGNElement* elements that describe specific segments of the GN database by their direction, identifier, location order, and relative truncation (for the first and last sections).

This is native means of location referencing by NDIC.

```

88     <globalNetworkLinear>
89       <networkVersion>
90         <values>
91           <value lang="cs">15.06</value>
92         </values>
93       </networkVersion>
94       <linearGeometryType>continuous</linearGeometryType>
95       <startPoint>
96         <sjtskPointCoordinates>
97           <sjtskX>-726024</sjtskX>
98           <sjtskY>-1062544</sjtskY>
99         </sjtskPointCoordinates>
100      </startPoint>
101      <endPoint>
102        <sjtskPointCoordinates>
103          <sjtskX>-727210</sjtskX>
104          <sjtskY>-1061636</sjtskY>
105        </sjtskPointCoordinates>
106      </endPoint>
107      <linearWithinLinearGNElement>
108        <sectionId>1687624</sectionId>
109        <directionRelativeOnLinearSection>opposite</
110      <directionRelativeOnLinearSection>
111        <orderOfSection>0</orderOfSection>
112        <fromPoint xsi:type="PercentageDistanceAlongLinearElement">
113          <percentageDistanceAlong>
114            0.349030470911379</percentageDistanceAlong>
115          </fromPoint>
116          <toPoint xsi:type="PercentageDistanceAlongLinearElement">
117            <percentageDistanceAlong>1</percentageDistanceAlong>
118          </toPoint>
119        </linearWithinLinearGNElement>
120        <linearWithinLinearGNElement>
121          <sectionId>1601936</sectionId>
122          <directionRelativeOnLinearSection>opposite</
123      <directionRelativeOnLinearSection>
124        <orderOfSection>1</orderOfSection>
125        <fromPoint xsi:type="PercentageDistanceAlongLinearElement">
126          <percentageDistanceAlong>0</percentageDistanceAlong>
127          </fromPoint>
128          <toPoint xsi:type="PercentageDistanceAlongLinearElement">
129            <percentageDistanceAlong>1</percentageDistanceAlong>
130          </toPoint>
131        </linearWithinLinearGNElement>
132        <linearWithinLinearGNElement>
133          <sectionId>3068685</sectionId>
134          <directionRelativeOnLinearSection>opposite</
135      <directionRelativeOnLinearSection>
136        <orderOfSection>2</orderOfSection>
137        <fromPoint xsi:type="PercentageDistanceAlongLinearElement">

```

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```

135         <percentageDistanceAlong>0</percentageDistanceAlong>
136     </fromPoint>
137     <toPoint xsi:type="PercentageDistanceAlongLinearElement">
138         <percentageDistanceAlong>1</percentageDistanceAlong>
139     </toPoint>
140 </linearWithinLinearGNElement>
141 <linearWithinLinearGNElement>
142     <sectionId>3068686</sectionId>
143     <directionRelativeOnLinearSection>opposite</
144     <directionRelativeOnLinearSection>
145     <orderOfSection>3</orderOfSection>
146     <fromPoint xsi:type="PercentageDistanceAlongLinearElement">
147         <percentageDistanceAlong>0</percentageDistanceAlong>
148     </fromPoint>
149     <toPoint xsi:type="PercentageDistanceAlongLinearElement">
150         <percentageDistanceAlong>
151         0.00793650792934262</percentageDistanceAlong>
152     </toPoint>
153 </linearWithinLinearGNElement>
</globalNetworkLinear>

```

The method for description of a point location *pointWithinLinearGNElement* only describes one distance from the beginning of the identified segment of GN database instead of two distances. Is very similar to the *pointAlongLinearElement* method.

## 2.5.4 Location description using Coordinates

Point location description without a reference to the linear is not allowed.

Both methods *pointByCoordinates* and *pointByMultiCoordinates* could be used only within the framework of the above defined methods.

The point location is described using geographic coordinates. Two methods may be used based on used coordinate reference system.

Method *pointByCoordinates* is used for description of position in WGS-84 coordinate system.

Method *pointByMultiCoordinates* is used to describe location by one or more coordinate systems, one of which being Czech system S-JTSK and second WGS-84. This is a DATEX II model extension type B.

```

95     <startPoint>
96         <sjtskPointCoordinates>
97             <sjtskX>-726024</sjtskX>
98             <sjtskY>-1062544</sjtskY>
99         </sjtskPointCoordinates>
100     </startPoint>

```



## APPENDIX A: MESSAGE EXAMPLES

### 3.1 restriction.xml - road closed for vehicles heavier than 3.5 tons

Publication contains exactly one message about road closure for vehicles heavier than 3.5 tons.

```
<?xml version="1.0" encoding="utf-8"?>
<d2LogicalModel modelBaseVersion="2"
xmlns="http://datex2.eu/schema/2/2_0"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://datex2.eu/schema/2/2_0 ../schema/schema.xsd">

  <exchange>
    <supplierIdentification>
      <country>cz</country>
      <nationalIdentifier>NDIC</nationalIdentifier>
    </supplierIdentification>
  </exchange>
  <payloadPublication xsi:type="SituationPublication" lang="cz">
    <feedType>x-format:cz-ndic_d2-restrictions-v1.1</feedType>
    <publicationTime>2017-07-01T23:11:30+02:00</publicationTime>
    <publicationCreator>
      <country>cz</country>
      <nationalIdentifier>NDIC</nationalIdentifier>
    </publicationCreator>
    <situation id="ff808181-49e6-e823-014a-07e78fe011e5" version="2">
      <headerInformation>
        <confidentiality>noRestriction</confidentiality>
        <informationStatus>real</informationStatus>
        <urgency>normalUrgency</urgency>
      </headerInformation>
      <situationRecord xsi:type="RoadOrCarriagewayOrLaneManagement"
id="ff808181-49e6-e823-014a-07e78fe011e5" version="1">
        <situationRecordCreationTime>2017-07-01T23:05:00+02:00</
↪situationRecordCreationTime>
        <situationRecordVersionTime>2017-07-01T23:11:30+02:00</
↪situationRecordVersionTime>
        <probabilityOfOccurrence>certain</probabilityOfOccurrence>
        <source>
          <sourceIdentification>NDIC</sourceIdentification>
          <sourceName>
            <values>
              <value lang="cs">Operátoři NDIC</value>
            </values>
          </sourceName>
        </source>
```

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```

<validity>
  <validityStatus>definedByValidityTimeSpec</validityStatus>
  <validityTimeSpecification>
    <overallStartTime>2017-07-01T23:05:00+02:00</overallStartTime>
    <overallEndTime>2017-07-02T10:00:00+02:00</overallEndTime>
  </validityTimeSpecification>
</validity>
<impact>
  <capacityRemaining>0</capacityRemaining>
</impact>
<generalPublicComment>
  <comment>
    <values>
      <value lang="cs">Uzavírka komunikace pro vozidla nad 3.5
        t</value>
    </values>
  </comment>
</generalPublicComment>
<groupOfLocations xsi:type="Linear">
  <alertCLinear xsi:type="AlertCMethod2Linear">
    <alertCLocationCountryCode>CZ</alertCLocationCountryCode>
    <alertCLocationTableNumber>25</alertCLocationTableNumber>
    <alertCLocationTableVersion>6.0</alertCLocationTableVersion>
    <alertCDirection>
      <alertCDirectionCoded>negative</alertCDirectionCoded>
    </alertCDirection>
    <alertCMethod2PrimaryPointLocation>
      <alertCLocation>
        <specificLocation>25532</specificLocation>
      </alertCLocation>
    </alertCMethod2PrimaryPointLocation>
    <alertCMethod2SecondaryPointLocation>
      <alertCLocation>
        <specificLocation>25531</specificLocation>
      </alertCLocation>
    </alertCMethod2SecondaryPointLocation>
  </alertCLinear>
  <linearWithinLinearElement>
    <directionRelativeOnLinearSection>opposite</
    directionRelativeOnLinearSection>
    <linearElement>
      <roadNumber>D1</roadNumber>
    </linearElement>
    <fromPoint xsi:type="DistanceFromLinearElementStart">
      <distanceAlong>18500</distanceAlong>
    </fromPoint>
    <toPoint xsi:type="DistanceFromLinearElementStart">
      <distanceAlong>17000</distanceAlong>
    </toPoint>
  </linearWithinLinearElement>
</globalNetworkLinear>
  <networkVersion>
    <values>
      <value lang="cs">15.06</value>
    </values>
  </networkVersion>

```

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```

<linearGeometryType>continuous</linearGeometryType>
<startPoint>
  <sjtskPointCoordinates>
    <sjtskX>-726024</sjtskX>
    <sjtskY>-1062544</sjtskY>
  </sjtskPointCoordinates>
</startPoint>
<endPoint>
  <sjtskPointCoordinates>
    <sjtskX>-727210</sjtskX>
    <sjtskY>-1061636</sjtskY>
  </sjtskPointCoordinates>
</endPoint>
<linearWithinLinearGNElement>
  <sectionId>1687624</sectionId>
  <directionRelativeOnLinearSection>opposite</
↪directionRelativeOnLinearSection>
  <orderOfSection>0</orderOfSection>
  <fromPoint xsi:type="PercentageDistanceAlongLinearElement">
    <percentageDistanceAlong>
      0.349030470911379</percentageDistanceAlong>
    </fromPoint>
  <toPoint xsi:type="PercentageDistanceAlongLinearElement">
    <percentageDistanceAlong>1</percentageDistanceAlong>
  </toPoint>
</linearWithinLinearGNElement>
<linearWithinLinearGNElement>
  <sectionId>1601936</sectionId>
  <directionRelativeOnLinearSection>opposite</
↪directionRelativeOnLinearSection>
  <orderOfSection>1</orderOfSection>
  <fromPoint xsi:type="PercentageDistanceAlongLinearElement">
    <percentageDistanceAlong>0</percentageDistanceAlong>
  </fromPoint>
  <toPoint xsi:type="PercentageDistanceAlongLinearElement">
    <percentageDistanceAlong>1</percentageDistanceAlong>
  </toPoint>
</linearWithinLinearGNElement>
<linearWithinLinearGNElement>
  <sectionId>3068685</sectionId>
  <directionRelativeOnLinearSection>opposite</
↪directionRelativeOnLinearSection>
  <orderOfSection>2</orderOfSection>
  <fromPoint xsi:type="PercentageDistanceAlongLinearElement">
    <percentageDistanceAlong>0</percentageDistanceAlong>
  </fromPoint>
  <toPoint xsi:type="PercentageDistanceAlongLinearElement">
    <percentageDistanceAlong>1</percentageDistanceAlong>
  </toPoint>
</linearWithinLinearGNElement>
<linearWithinLinearGNElement>
  <sectionId>3068686</sectionId>
  <directionRelativeOnLinearSection>opposite</
↪directionRelativeOnLinearSection>
  <orderOfSection>3</orderOfSection>
  <fromPoint xsi:type="PercentageDistanceAlongLinearElement">

```

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```

        <percentageDistanceAlong>0</percentageDistanceAlong>
      </fromPoint>
      <toPoint xsi:type="PercentageDistanceAlongLinearElement">
        <percentageDistanceAlong>
          0.00793650792934262</percentageDistanceAlong>
        </toPoint>
      </linearWithinLinearGNElement>
    </globalNetworkLinear>
  </groupOfLocations>
  <complianceOption>mandatory</complianceOption>
  <forVehiclesWithCharacteristicsOf>
    <vehicleType>lorry</vehicleType>
    <grossWeightCharacteristic>
      <comparisonOperator>greaterThan</comparisonOperator>
      <grossVehicleWeight>3.5</grossVehicleWeight>
    </grossWeightCharacteristic>
  </forVehiclesWithCharacteristicsOf>
  <roadOrCarriagewayOrLaneManagementType>roadClosed</
↪ roadOrCarriagewayOrLaneManagementType>
</situationRecord>
</situation>
</payloadPublication>
</d2LogicalModel>

```

### 3.1.1 Root element and publication envelope

Root element *d2LogicalModel* declares, that this message is using DATEX II.

Element *exchange* contains provider identifier.

Element *payloadPublication* is an envelope for actual information payload containing general information and one or more elements *situation*, (in our case there is one).

Attribute *xsi:type="SituationPublication"* in element *payloadPublication* determines, that this envelope is of type Situation publication.

Element *feedType* states publication schema by its uri.

Element *publicationTime* states publication time of this publication (file) as a whole.

Element *publicationCreator* identifies creator of the publication.

### 3.1.2 Description of one specific situation

Element *situation* describes a particular event, more particular events means more *situation* elements. Each *situation* element may contain one or more individual sub-messages, elements *situationRecord*. In this case occurs in the element *situation* just once.

Element *situationVersionTime* sets the time of the publication of this version of the situation.

Element *headerInformation* describes how to handle the information described.

### Event description (what happened)

Attribute *xsi:type="RoadOrCarriagewayOrLaneManagement"* of element *situationRecord* defines event type. In DATEX II model [d2ref] it can be verified, that given attribute is of type descendant of class *OperatorActions* in following descendant sequence *RoadOrCarriagewayOrLaneManagement* -> *NetworkManagement* -> *OperatorActions*.

Element *complianceOption* with value *mandatory* states that the restriction is mandatory.

Element *forVehiclesWithCharacteristicsOf* specifies by value *lorry* in *vehicleType* that only heavy vehicles are affected by restriction and *grossWeightCharacteristic* states that only vehicles heavier than 3.5 tons are affected.

Element *roadOrCarriagewayOrLaneManagementType* with value *roadClosed* states that the road is closed for given vehicle types.

### Time validity

Estimated time validity is given in a structure *validity* by values of validity interval start *overallStartTime* and validity interval end *overallEndTime*.

### Event location

Event location is described in element *groupOfLocations* directly by embedded description. Reference to predefined locations is not used. Attribute *xsi:type="Linear"* states, that given element is of type linear location, which can be expressed by different location referencing methods.

The position of linear type is represented by three descriptions. First, using element *alertCLinear* of type *AlertCMethod2Linear*, then using a general description of the position on the road in element *linearWithinLinearElement* and then described using Global Network links in element *globalNetworkLinear*.

All location referencing methods used are describing the same location.

### Additional content

The document states time of its first version in *situationRecordCreationTime* and time of current version in *situationRecordVersionTime* for each *situationRecord*.

Element *source* lists the the creator of this information, who is responsible for its accuracy.

## APPENDIX B: W3C XML SCHEMA

XML structure of a transmitted message is defined by following W3C XML schema:

### 4.1 *schema.xsd*

```

1 <?xml version="1.0" encoding="utf-8" standalone="no"?>
2 <xs:schema elementFormDefault="qualified" attributeFormDefault="unqualified"
  ↳ xmlns:D2LogicalModel="http://datex2.eu/schema/2/2_0" version="2.3" targetNamespace=
  ↳ "http://datex2.eu/schema/2/2_0" xmlns:xs="http://www.w3.org/2001/XMLSchema">
3   <xs:complexType name="_ExtensionType">
4     <xs:sequence>
5       <xs:any namespace="##any" processContents="lax" minOccurs="0" maxOccurs=
  ↳ "unbounded" />
6     </xs:sequence>
7   </xs:complexType>
8   <xs:complexType name="_IntermediatePointOnLinearElement">
9     <xs:sequence>
10      <xs:element name="referent" type="D2LogicalModel:Referent" minOccurs="1"
  ↳ maxOccurs="1" />
11    </xs:sequence>
12    <xs:attribute name="index" type="xs:int" use="required" />
13  </xs:complexType>
14  <xs:complexType name="_LinearExtensionType">
15    <xs:sequence>
16      <xs:element name="openlrExtendedLinear" type=
  ↳ "D2LogicalModel:OpenlrExtendedLinear" minOccurs="0" />
17      <xs:any namespace="##other" processContents="lax" minOccurs="0" maxOccurs=
  ↳ "unbounded" />
18    </xs:sequence>
19  </xs:complexType>
20  <xs:complexType name="_PointExtensionType">
21    <xs:sequence>
22      <xs:element name="openlrExtendedPoint" type="D2LogicalModel:OpenlrExtendedPoint
  ↳ " minOccurs="0" />
23      <xs:any namespace="##other" processContents="lax" minOccurs="0" maxOccurs=
  ↳ "unbounded" />
24    </xs:sequence>
25  </xs:complexType>
26  <xs:complexType name="AffectedCarriagewayAndLanes">
27    <xs:sequence>
28      <xs:element name="carriageway" type="D2LogicalModel:CarriagewayEnum" minOccurs=
  ↳ "1" maxOccurs="1" />
29      <xs:element name="lane" type="D2LogicalModel:LaneEnum" minOccurs="0" maxOccurs=
  ↳ "unbounded" />

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```

30     <xs:element name="footpath" type="D2LogicalModel:Boolean" minOccurs="0"
↪ maxOccurs="1" />
31     <xs:element name="lengthAffected" type="D2LogicalModel:MetresAsFloat" minOccurs=
↪ "0" maxOccurs="1" />
32     <xs:element name="affectedCarriagewayAndLanesExtension" type="D2LogicalModel:_
↪ ExtensionType" minOccurs="0" />
33     </xs:sequence>
34 </xs:complexType>
35 <xs:complexType name="AlertCDirection">
36     <xs:sequence>
37         <xs:element name="alertCDirectionCoded" type="D2LogicalModel:AlertCDirectionEnum
↪ " minOccurs="1" maxOccurs="1" />
38         <xs:element name="alertCDirectionNamed" type="D2LogicalModel:MultilingualString
↪ " minOccurs="0" maxOccurs="1" />
39         <xs:element name="alertCDirectionSense" type="D2LogicalModel:Boolean" minOccurs=
↪ "0" maxOccurs="1" />
40         <xs:element name="alertCDirectionExtension" type="D2LogicalModel:_ExtensionType
↪ " minOccurs="0" />
41     </xs:sequence>
42 </xs:complexType>
43 <xs:simpleType name="AlertCDirectionEnum">
44     <xs:restriction base="xs:string">
45         <xs:enumeration value="both" />
46         <xs:enumeration value="negative" />
47         <xs:enumeration value="positive" />
48         <xs:enumeration value="unknown" />
49     </xs:restriction>
50 </xs:simpleType>
51 <xs:complexType name="AlertCLinear" abstract="true">
52     <xs:sequence>
53         <xs:element name="alertCLocationCountryCode" type="D2LogicalModel:String"
↪ minOccurs="1" maxOccurs="1" />
54         <xs:element name="alertCLocationTableNumber" type="D2LogicalModel:String"
↪ minOccurs="1" maxOccurs="1" />
55         <xs:element name="alertCLocationTableVersion" type="D2LogicalModel:String"
↪ minOccurs="1" maxOccurs="1" />
56         <xs:element name="alertCLinearExtension" type="D2LogicalModel:_ExtensionType"
↪ minOccurs="0" />
57     </xs:sequence>
58 </xs:complexType>
59 <xs:complexType name="AlertCLinearByCode">
60     <xs:complexContent>
61         <xs:extension base="D2LogicalModel:AlertCLinear">
62             <xs:sequence>
63                 <xs:element name="alertCDirection" type="D2LogicalModel:AlertCDirection" />
64                 <xs:element name="locationCodeForLinearLocation" type=
↪ "D2LogicalModel:AlertCLocation" />
65                 <xs:element name="alertCLinearByCodeExtension" type="D2LogicalModel:_
↪ ExtensionType" minOccurs="0" />
66             </xs:sequence>
67         </xs:extension>
68     </xs:complexContent>
69 </xs:complexType>
70 <xs:complexType name="AlertCLocation">
71     <xs:sequence>
72         <xs:element name="alertCLocationName" type="D2LogicalModel:MultilingualString"
↪

```

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```

73   <minOccurs="0" maxOccurs="1" />
      <xs:element name="specificLocation" type="D2LogicalModel:AlertCLocationCode"
74   <minOccurs="1" maxOccurs="1" />
      <xs:element name="alertCLocationExtension" type="D2LogicalModel:_ExtensionType"
75   <minOccurs="0" />
      </xs:sequence>
76   </xs:complexType>
77   <xs:simpleType name="AlertCLocationCode">
78     <xs:restriction base="D2LogicalModel:NonNegativeInteger" />
79   </xs:simpleType>
80   <xs:complexType name="AlertCMethod2Linear">
81     <xs:complexContent>
82       <xs:extension base="D2LogicalModel:AlertCLinear">
83         <xs:sequence>
84           <xs:element name="alertCDirection" type="D2LogicalModel:AlertCDirection" />
85           <xs:element name="alertCMethod2PrimaryPointLocation" type=
86             "D2LogicalModel:AlertCMethod2PrimaryPointLocation" />
87           <xs:element name="alertCMethod2SecondaryPointLocation" type=
88             "D2LogicalModel:AlertCMethod2SecondaryPointLocation" />
89           <xs:element name="alertCMethod2LinearExtension" type="D2LogicalModel:_
90             ExtensionType" minOccurs="0" />
91         </xs:sequence>
92       </xs:extension>
93     </xs:complexContent>
94   </xs:complexType>
95   <xs:complexType name="AlertCMethod2Point">
96     <xs:complexContent>
97       <xs:extension base="D2LogicalModel:AlertCPoint">
98         <xs:sequence>
99           <xs:element name="alertCDirection" type="D2LogicalModel:AlertCDirection" />
100          <xs:element name="alertCMethod2PrimaryPointLocation" type=
101            "D2LogicalModel:AlertCMethod2PrimaryPointLocation" />
102          <xs:element name="alertCMethod2PointExtension" type="D2LogicalModel:_
103            ExtensionType" minOccurs="0" />
104        </xs:sequence>
105      </xs:extension>
106    </xs:complexContent>
107  </xs:complexType>
108  <xs:complexType name="AlertCMethod2PrimaryPointLocation">
109    <xs:sequence>
110      <xs:element name="alertCLocation" type="D2LogicalModel:AlertCLocation" />
111      <xs:element name="alertCMethod2PrimaryPointLocationExtension" type=
112        "D2LogicalModel:_ExtensionType" minOccurs="0" />
113    </xs:sequence>
114  </xs:complexType>
115  <xs:complexType name="AlertCMethod2SecondaryPointLocation">
116    <xs:sequence>
117      <xs:element name="alertCLocation" type="D2LogicalModel:AlertCLocation" />
118      <xs:element name="alertCMethod2SecondaryPointLocationExtension" type=
119        "D2LogicalModel:_ExtensionType" minOccurs="0" />
120    </xs:sequence>
121  </xs:complexType>
122  <xs:complexType name="AlertCPoint" abstract="true">
123    <xs:sequence>
124      <xs:element name="alertCLocationCountryCode" type="D2LogicalModel:String"
125        <minOccurs="1" maxOccurs="1" />

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```

118     <xs:element name="alertCLocationTableNumber" type="D2LogicalModel:String"
↪minOccurs="1" maxOccurs="1" />
119     <xs:element name="alertCLocationTableVersion" type="D2LogicalModel:String"
↪minOccurs="1" maxOccurs="1" />
120     <xs:element name="alertCPointExtension" type="D2LogicalModel:_ExtensionType"
↪minOccurs="0" />
121   </xs:sequence>
122 </xs:complexType>
123 <xs:simpleType name="AngleInDegrees">
124   <xs:restriction base="D2LogicalModel:NonNegativeInteger" />
125 </xs:simpleType>
126 <xs:complexType name="Area">
127   <xs:complexContent>
128     <xs:extension base="D2LogicalModel:Location">
129       <xs:sequence>
130         <xs:element name="areaExtension" type="D2LogicalModel:_ExtensionType"
↪minOccurs="0" />
131       </xs:sequence>
132     </xs:extension>
133   </xs:complexContent>
134 </xs:complexType>
135 <xs:complexType name="AreaDestination">
136   <xs:complexContent>
137     <xs:extension base="D2LogicalModel:Destination">
138       <xs:sequence>
139         <xs:element name="area" type="D2LogicalModel:Area" />
140         <xs:element name="areaDestinationExtension" type="D2LogicalModel:_
↪ExtensionType" minOccurs="0" />
141       </xs:sequence>
142     </xs:extension>
143   </xs:complexContent>
144 </xs:complexType>
145 <xs:complexType name="AxleFlowValue">
146   <xs:complexContent>
147     <xs:extension base="D2LogicalModel:DataValue">
148       <xs:sequence>
149         <xs:element name="axleFlowRate" type="D2LogicalModel:AxlesPerHour"
↪minOccurs="1" maxOccurs="1" />
150         <xs:element name="axleFlowValueExtension" type="D2LogicalModel:_
↪ExtensionType" minOccurs="0" />
151       </xs:sequence>
152     </xs:extension>
153   </xs:complexContent>
154 </xs:complexType>
155 <xs:simpleType name="AxlesPerHour">
156   <xs:restriction base="D2LogicalModel:NonNegativeInteger" />
157 </xs:simpleType>
158 <xs:simpleType name="Boolean">
159   <xs:restriction base="xs:boolean" />
160 </xs:simpleType>
161 <xs:simpleType name="CarriagewayEnum">
162   <xs:restriction base="xs:string">
163     <xs:enumeration value="connectingCarriageway" />
164     <xs:enumeration value="entrySlipRoad" />
165     <xs:enumeration value="exitSlipRoad" />
166     <xs:enumeration value="flyover" />

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```

167     <xs:enumeration value="leftHandFeederRoad" />
168     <xs:enumeration value="leftHandParallelCarriageway" />
169     <xs:enumeration value="mainCarriageway" />
170     <xs:enumeration value="oppositeCarriageway" />
171     <xs:enumeration value="parallelCarriageway" />
172     <xs:enumeration value="rightHandFeederRoad" />
173     <xs:enumeration value="rightHandParallelCarriageway" />
174     <xs:enumeration value="roundabout" />
175     <xs:enumeration value="serviceRoad" />
176     <xs:enumeration value="slipRoads" />
177     <xs:enumeration value="underpass" />
178   </xs:restriction>
179 </xs:simpleType>
180 <xs:complexType name="Cause" abstract="true">
181   <xs:sequence>
182     <xs:element name="causeExtension" type="D2LogicalModel:_ExtensionType"
183     ↪ minOccurs="0" />
184   </xs:sequence>
185 </xs:complexType>
186 <xs:simpleType name="CauseTypeEnum">
187   <xs:restriction base="xs:string">
188     <xs:enumeration value="accident" />
189     <xs:enumeration value="congestion" />
190     <xs:enumeration value="earlierAccident" />
191     <xs:enumeration value="earlierEvent" />
192     <xs:enumeration value="earlierIncident" />
193     <xs:enumeration value="equipmentFailure" />
194     <xs:enumeration value="excessiveHeat" />
195     <xs:enumeration value="frost" />
196     <xs:enumeration value="holidayTraffic" />
197     <xs:enumeration value="infrastructureFailure" />
198     <xs:enumeration value="largeNumbersOfVisitors" />
199     <xs:enumeration value="obstruction" />
200     <xs:enumeration value="pollutionAlert" />
201     <xs:enumeration value="poorWeather" />
202     <xs:enumeration value="problemsAtBorderPost" />
203     <xs:enumeration value="problemsAtCustomPost" />
204     <xs:enumeration value="problemsOnLocalRoads" />
205     <xs:enumeration value="radioactiveLeakAlert" />
206     <xs:enumeration value="roadsideEvent" />
207     <xs:enumeration value="rubberNecking" />
208     <xs:enumeration value="securityIncident" />
209     <xs:enumeration value="shearWeightOfTraffic" />
210     <xs:enumeration value="technicalProblems" />
211     <xs:enumeration value="terrorism" />
212     <xs:enumeration value="toxicCloudAlert" />
213     <xs:enumeration value="vandalism" />
214     <xs:enumeration value="other" />
215   </xs:restriction>
216 </xs:simpleType>
217 <xs:complexType name="Comment">
218   <xs:sequence>
219     <xs:element name="comment" type="D2LogicalModel:MultilingualString" minOccurs="1
220     ↪ maxOccurs="1" />
221     <xs:element name="commentDateTime" type="D2LogicalModel:DateTime" minOccurs="0"
222     ↪ maxOccurs="1" />

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```

220   <xs:element name="commentType" type="D2LogicalModel:CommentTypeEnum" minOccurs=
↪ "0" maxOccurs="1" />
221   <xs:element name="commentExtension" type="D2LogicalModel:_ExtensionType"
↪ minOccurs="0" />
222   </xs:sequence>
223 </xs:complexType>
224 <xs:simpleType name="CommentTypeEnum">
225   <xs:restriction base="xs:string">
226     <xs:enumeration value="abnormalLoadMovementNote" />
227     <xs:enumeration value="dataProcessingNote" />
228     <xs:enumeration value="description" />
229     <xs:enumeration value="internalNote" />
230     <xs:enumeration value="locationDescriptor" />
231     <xs:enumeration value="warning" />
232     <xs:enumeration value="other" />
233   </xs:restriction>
234 </xs:simpleType>
235 <xs:simpleType name="ComparisonOperatorEnum">
236   <xs:restriction base="xs:string">
237     <xs:enumeration value="equalTo" />
238     <xs:enumeration value="greaterThan" />
239     <xs:enumeration value="greaterThanOrEqualTo" />
240     <xs:enumeration value="lessThan" />
241     <xs:enumeration value="lessThanOrEqualTo" />
242   </xs:restriction>
243 </xs:simpleType>
244 <xs:simpleType name="ComplianceOptionEnum">
245   <xs:restriction base="xs:string">
246     <xs:enumeration value="advisory" />
247     <xs:enumeration value="mandatory" />
248   </xs:restriction>
249 </xs:simpleType>
250 <xs:simpleType name="ComputationMethodEnum">
251   <xs:restriction base="xs:string">
252     <xs:enumeration value="arithmeticAverageOfSamplesBasedOnAFixedNumberOfSamples" /
↪ >
253     <xs:enumeration value="arithmeticAverageOfSamplesInATimePeriod" />
254     <xs:enumeration value="harmonicAverageOfSamplesInATimePeriod" />
255     <xs:enumeration value="medianOfSamplesInATimePeriod" />
256     <xs:enumeration value="movingAverageOfSamples" />
257   </xs:restriction>
258 </xs:simpleType>
259 <xs:complexType name="ConcentrationOfVehiclesValue">
260   <xs:complexContent>
261     <xs:extension base="D2LogicalModel:DataValue">
262       <xs:sequence>
263         <xs:element name="concentrationOfVehicles" type=
↪ "D2LogicalModel:ConcentrationVehiclesPerKilometre" minOccurs="1" maxOccurs="1" />
264         <xs:element name="concentrationOfVehiclesValueExtension" type=
↪ "D2LogicalModel:_ExtensionType" minOccurs="0" />
265       </xs:sequence>
266     </xs:extension>
267   </xs:complexContent>
268 </xs:complexType>
269 <xs:simpleType name="ConcentrationVehiclesPerKilometre">
270   <xs:restriction base="D2LogicalModel:NonNegativeInteger" />

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271 </xs:simpleType>
272 <xs:simpleType name="ConfidentialityValueEnum">
273   <xs:restriction base="xs:string">
274     <xs:enumeration value="internalUse" />
275     <xs:enumeration value="noRestriction" />
276     <xs:enumeration value="restrictedToAuthorities" />
277     <xs:enumeration value="restrictedToAuthoritiesAndTrafficOperators" />
278     <xs:enumeration value="restrictedToAuthoritiesTrafficOperatorsAndPublishers" />
279     <xs:enumeration value="restrictedToAuthoritiesTrafficOperatorsAndVms" />
280   </xs:restriction>
281 </xs:simpleType>
282 <xs:complexType name="ConstructionWorks">
283   <xs:complexContent>
284     <xs:extension base="D2LogicalModel:Roadworks">
285       <xs:sequence>
286         <xs:element name="constructionWorkType" type=
287 ↪ "D2LogicalModel:ConstructionWorkTypeEnum" minOccurs="0" maxOccurs="1" />
288         <xs:element name="constructionWorksExtension" type="D2LogicalModel:_
289 ↪ ExtensionType" minOccurs="0" />
290       </xs:sequence>
291     </xs:extension>
292   </xs:complexContent>
293 </xs:complexType>
294 <xs:simpleType name="ConstructionWorkTypeEnum">
295   <xs:restriction base="xs:string">
296     <xs:enumeration value="blastingWork" />
297     <xs:enumeration value="constructionWork" />
298     <xs:enumeration value="demolitionWork" />
299     <xs:enumeration value="roadImprovementOrUpgrading" />
300     <xs:enumeration value="roadWideningWork" />
301   </xs:restriction>
302 </xs:simpleType>
303 <xs:simpleType name="CountryEnum">
304   <xs:restriction base="xs:string">
305     <xs:enumeration value="at" />
306     <xs:enumeration value="be" />
307     <xs:enumeration value="bg" />
308     <xs:enumeration value="ch" />
309     <xs:enumeration value="cs" />
310     <xs:enumeration value="cy" />
311     <xs:enumeration value="cz" />
312     <xs:enumeration value="de" />
313     <xs:enumeration value="dk" />
314     <xs:enumeration value="ee" />
315     <xs:enumeration value="es" />
316     <xs:enumeration value="fi" />
317     <xs:enumeration value="fo" />
318     <xs:enumeration value="fr" />
319     <xs:enumeration value="gb" />
320     <xs:enumeration value="gg" />
321     <xs:enumeration value="gi" />
322     <xs:enumeration value="gr" />
323     <xs:enumeration value="hr" />
324     <xs:enumeration value="hu" />
325     <xs:enumeration value="ie" />
326     <xs:enumeration value="im" />

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325     <xs:enumeration value="is" />
326     <xs:enumeration value="it" />
327     <xs:enumeration value="je" />
328     <xs:enumeration value="li" />
329     <xs:enumeration value="lt" />
330     <xs:enumeration value="lu" />
331     <xs:enumeration value="lv" />
332     <xs:enumeration value="ma" />
333     <xs:enumeration value="mc" />
334     <xs:enumeration value="mk" />
335     <xs:enumeration value="mt" />
336     <xs:enumeration value="nl" />
337     <xs:enumeration value="no" />
338     <xs:enumeration value="pl" />
339     <xs:enumeration value="pt" />
340     <xs:enumeration value="ro" />
341     <xs:enumeration value="se" />
342     <xs:enumeration value="si" />
343     <xs:enumeration value="sk" />
344     <xs:enumeration value="sm" />
345     <xs:enumeration value="tr" />
346     <xs:enumeration value="va" />
347     <xs:enumeration value="other" />
348   </xs:restriction>
349 </xs:simpleType>
350 <xs:element name="d2LogicalModel" type="D2LogicalModel:D2LogicalModel">
351   <xs:unique name="_d2LogicalModelSituationRecordConstraint">
352     <xs:selector xpath="."/D2LogicalModel:situationRecord" />
353     <xs:field xpath="@id" />
354     <xs:field xpath="@version" />
355   </xs:unique>
356   <xs:unique name="_d2LogicalModelSituationConstraint">
357     <xs:selector xpath="."/D2LogicalModel:situation" />
358     <xs:field xpath="@id" />
359     <xs:field xpath="@version" />
360   </xs:unique>
361 </xs:element>
362 <xs:complexType name="D2LogicalModel">
363   <xs:sequence>
364     <xs:element name="exchange" type="D2LogicalModel:Exchange" />
365     <xs:element name="payloadPublication" type="D2LogicalModel:PayloadPublication"
366     ↪ minOccurs="0" />
367     <xs:element name="d2LogicalModelExtension" type="D2LogicalModel:_ExtensionType"
368     ↪ minOccurs="0" />
369   </xs:sequence>
370   <xs:attribute name="modelBaseVersion" use="required" fixed="2" />
371 </xs:complexType>
372 <xs:complexType name="DataValue" abstract="true">
373   <xs:sequence>
374     <xs:element name="dataError" type="D2LogicalModel:Boolean" minOccurs="0"
375     ↪ maxOccurs="1" />
376     <xs:element name="reasonForDataError" type="D2LogicalModel:MultilingualString"
377     ↪ minOccurs="0" maxOccurs="1" />
378     <xs:element name="dataValueExtension" type="D2LogicalModel:_ExtensionType"
379     ↪ minOccurs="0" />
380   </xs:sequence>

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376     <xs:attribute name="accuracy" type="D2LogicalModel:Percentage" use="optional" />
377     <xs:attribute name="computationalMethod" type=
378 ↪ "D2LogicalModel:ComputationMethodEnum" use="optional" />
379     <xs:attribute name="numberOfIncompleteInputs" type=
380 ↪ "D2LogicalModel:NonNegativeInteger" use="optional" />
381     <xs:attribute name="numberOfInputValuesUsed" type=
382 ↪ "D2LogicalModel:NonNegativeInteger" use="optional" />
383     <xs:attribute name="smoothingFactor" type="D2LogicalModel:Float" use="optional" />
384     <xs:attribute name="standardDeviation" type="D2LogicalModel:Float" use="optional" ↪
385 ↪ />
386     <xs:attribute name="supplierCalculatedDataQuality" type="D2LogicalModel:Percentage
387 ↪ " use="optional" />
388 </xs:complexType>
389 <xs:simpleType name="DateTime">
390     <xs:restriction base="xs:dateTime" />
391 </xs:simpleType>
392 <xs:complexType name="DateTimeValue">
393     <xs:complexContent>
394         <xs:extension base="D2LogicalModel:DataValue">
395             <xs:sequence>
396                 <xs:element name="dateTime" type="D2LogicalModel:DateTime" minOccurs="1" ↪
397 ↪ maxOccurs="1" />
398                 <xs:element name="dateTimeValueExtension" type="D2LogicalModel:_
399 ↪ ExtensionType" minOccurs="0" />
400             </xs:sequence>
401         </xs:extension>
402     </xs:complexContent>
403 </xs:complexType>
404 <xs:complexType name="Delays">
405     <xs:sequence>
406         <xs:element name="delaysType" type="D2LogicalModel:DelaysTypeEnum" minOccurs="0
407 ↪ " maxOccurs="1" />
408         <xs:element name="delayTimeValue" type="D2LogicalModel:Seconds" minOccurs="0" ↪
409 ↪ maxOccurs="1" />
410         <xs:element name="delaysExtension" type="D2LogicalModel:_ExtensionType" ↪
411 ↪ minOccurs="0" />
412     </xs:sequence>
413 </xs:complexType>
414 <xs:simpleType name="DelaysTypeEnum">
415     <xs:restriction base="xs:string">
416         <xs:enumeration value="delays" />
417         <xs:enumeration value="delaysOfUncertainDuration" />
418         <xs:enumeration value="longDelays" />
419         <xs:enumeration value="veryLongDelays" />
420     </xs:restriction>
</xs:simpleType>
<xs:complexType name="Destination" abstract="true">
    <xs:sequence>
        <xs:element name="destinationExtension" type="D2LogicalModel:_ExtensionType" ↪
↪ minOccurs="0" />
    </xs:sequence>
</xs:complexType>
<xs:simpleType name="DirectionEnum">
    <xs:restriction base="xs:string">
        <xs:enumeration value="allDirections" />
        <xs:enumeration value="bothWays" />

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421     <xs:enumeration value="clockwise" />
422     <xs:enumeration value="anticlockwise" />
423     <xs:enumeration value="innerRing" />
424     <xs:enumeration value="outerRing" />
425     <xs:enumeration value="northBound" />
426     <xs:enumeration value="northEastBound" />
427     <xs:enumeration value="eastBound" />
428     <xs:enumeration value="southEastBound" />
429     <xs:enumeration value="southBound" />
430     <xs:enumeration value="southWestBound" />
431     <xs:enumeration value="westBound" />
432     <xs:enumeration value="northWestBound" />
433     <xs:enumeration value="inboundTowardsTown" />
434     <xs:enumeration value="outboundFromTown" />
435     <xs:enumeration value="unknown" />
436     <xs:enumeration value="opposite" />
437     <xs:enumeration value="other" />
438   </xs:restriction>
439 </xs:simpleType>
440 <xs:complexType name="DistanceAlongLinearElement" abstract="true">
441   <xs:sequence>
442     <xs:element name="distanceAlongLinearElementExtension" type="D2LogicalModel:_
↪ ExtensionType" minOccurs="0" />
443   </xs:sequence>
444 </xs:complexType>
445 <xs:complexType name="DistanceFromLinearElementReferent">
446   <xs:complexContent>
447     <xs:extension base="D2LogicalModel:DistanceAlongLinearElement">
448       <xs:sequence>
449         <xs:element name="distanceAlong" type="D2LogicalModel:MetresAsFloat" ↪
↪ minOccurs="1" maxOccurs="1" />
450         <xs:element name="fromReferent" type="D2LogicalModel:Referent" />
451         <xs:element name="towardsReferent" type="D2LogicalModel:Referent" minOccurs=
↪ "0" />
452         <xs:element name="distanceFromLinearElementReferentExtension" type=
↪ "D2LogicalModel:_ExtensionType" minOccurs="0" />
453       </xs:sequence>
454     </xs:extension>
455   </xs:complexContent>
456 </xs:complexType>
457 <xs:complexType name="DistanceFromLinearElementStart">
458   <xs:complexContent>
459     <xs:extension base="D2LogicalModel:DistanceAlongLinearElement">
460       <xs:sequence>
461         <xs:element name="distanceAlong" type="D2LogicalModel:MetresAsFloat" ↪
↪ minOccurs="1" maxOccurs="1" />
462         <xs:element name="distanceFromLinearElementStartExtension" type=
↪ "D2LogicalModel:_ExtensionType" minOccurs="0" />
463       </xs:sequence>
464     </xs:extension>
465   </xs:complexContent>
466 </xs:complexType>
467 <xs:complexType name="ElaboratedDataFault">
468   <xs:complexContent>
469     <xs:extension base="D2LogicalModel:Fault">
470       <xs:sequence>

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471     <xs:element name="elaboratedDataFault" type=
↪ "D2LogicalModel:ElaboratedDataFaultEnum" minOccurs="1" maxOccurs="1" />
472     <xs:element name="elaboratedDataFaultExtension" type="D2LogicalModel:_
↪ ExtensionType" minOccurs="0" />
473     </xs:sequence>
474   </xs:extension>
475 </xs:complexContent>
476 </xs:complexType>
477 <xs:simpleType name="ElaboratedDataFaultEnum">
478   <xs:restriction base="xs:string">
479     <xs:enumeration value="intermittentDataValues" />
480     <xs:enumeration value="noDataValuesAvailable" />
481     <xs:enumeration value="spuriousUnreliableDataValues" />
482     <xs:enumeration value="unspecifiedOrUnknownFault" />
483     <xs:enumeration value="other" />
484   </xs:restriction>
485 </xs:simpleType>
486 <xs:complexType name="Exchange">
487   <xs:sequence>
488     <xs:element name="supplierIdentification" type=
↪ "D2LogicalModel:InternationalIdentifier" />
489     <xs:element name="exchangeExtension" type="D2LogicalModel:_ExtensionType"
↪ minOccurs="0" />
490   </xs:sequence>
491 </xs:complexType>
492 <xs:complexType name="Fault">
493   <xs:sequence>
494     <xs:element name="faultIdentifier" type="D2LogicalModel:String" minOccurs="0"
↪ maxOccurs="1" />
495     <xs:element name="faultDescription" type="D2LogicalModel:String" minOccurs="0"
↪ maxOccurs="1" />
496     <xs:element name="faultCreationTime" type="D2LogicalModel:DateTime" minOccurs="0"
↪ maxOccurs="1" />
497     <xs:element name="faultLastUpdateTime" type="D2LogicalModel:DateTime" minOccurs=
↪ "1" maxOccurs="1" />
498     <xs:element name="faultSeverity" type="D2LogicalModel:FaultSeverityEnum"
↪ minOccurs="0" maxOccurs="1" />
499     <xs:element name="faultExtension" type="D2LogicalModel:_ExtensionType"
↪ minOccurs="0" />
500   </xs:sequence>
501 </xs:complexType>
502 <xs:simpleType name="FaultSeverityEnum">
503   <xs:restriction base="xs:string">
504     <xs:enumeration value="low" />
505     <xs:enumeration value="medium" />
506     <xs:enumeration value="high" />
507     <xs:enumeration value="unknown" />
508   </xs:restriction>
509 </xs:simpleType>
510 <xs:simpleType name="Float">
511   <xs:restriction base="xs:float" />
512 </xs:simpleType>
513 <xs:complexType name="GeneralInstructionOrMessageToRoadUsers">
514   <xs:complexContent>
515     <xs:extension base="D2LogicalModel:NetworkManagement">
516       <xs:sequence>

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517     <xs:element name="generalInstructionToRoadUsersType" type=
518     ↪ "D2LogicalModel:GeneralInstructionToRoadUsersTypeEnum" minOccurs="0" maxOccurs="1" /
519     ↪ >
520     <xs:element name="generalMessageToRoadUsers" type=
521     ↪ "D2LogicalModel:MultilingualString" minOccurs="0" maxOccurs="1" />
522     <xs:element name="generalInstructionOrMessageToRoadUsersExtension" type=
523     ↪ "D2LogicalModel:_ExtensionType" minOccurs="0" />
524     </xs:sequence>
525     </xs:extension>
526     </xs:complexContent>
527     </xs:complexType>
528     <xs:simpleType name="GeneralInstructionToRoadUsersTypeEnum">
529     <xs:restriction base="xs:string">
530     <xs:enumeration value="allowEmergencyVehiclesToPass" />
531     <xs:enumeration value="approachWithCare" />
532     <xs:enumeration value="avoidTheArea" />
533     <xs:enumeration value="closeAllWindowsTurnOffHeaterAndVents" />
534     <xs:enumeration value="crossJunctionWithCare" />
535     <xs:enumeration value="doNotAllowUnnecessaryGaps" />
536     <xs:enumeration value="doNotLeaveYourVehicle" />
537     <xs:enumeration value="doNotThrowOutAnyBurningObjects" />
538     <xs:enumeration value="doNotUseNavigationSystems" />
539     <xs:enumeration value="driveCarefully" />
540     <xs:enumeration value="driveWithExtremeCaution" />
541     <xs:enumeration value="flashYourLights" />
542     <xs:enumeration value="followTheVehicleInFrontSmoothly" />
543     <xs:enumeration value="increaseNormalFollowingDistance" />
544     <xs:enumeration value="inEmergencyWaitForPatrolService" />
545     <xs:enumeration value="keepYourDistance" />
546     <xs:enumeration value="leaveYourVehicleProceedToNextSafePlace" />
547     <xs:enumeration value="noNakedFlames" />
548     <xs:enumeration value="noOvertaking" />
549     <xs:enumeration value="noSmoking" />
550     <xs:enumeration value="noStopping" />
551     <xs:enumeration value="noUturns" />
552     <xs:enumeration value="observeAmberAlert" />
553     <xs:enumeration value="observeSignals" />
554     <xs:enumeration value="observeSigns" />
555     <xs:enumeration value="onlyTravelIfAbsolutelyNecessary" />
556     <xs:enumeration value="overtakeWithCare" />
557     <xs:enumeration value="pullOverToTheEdgeOfTheRoadway" />
558     <xs:enumeration value="stopAtNextSafePlace" />
559     <xs:enumeration value="stopAtNextServiceArea" />
560     <xs:enumeration value="switchOffEngine" />
561     <xs:enumeration value="switchOffMobilePhonesAndTwoWayRadios" />
562     <xs:enumeration value="testYourBrakes" />
563     <xs:enumeration value="useBusService" />
564     <xs:enumeration value="useFogLights" />
565     <xs:enumeration value="useHazardWarningLights" />
566     <xs:enumeration value="useHeadlights" />
567     <xs:enumeration value="useRailService" />
568     <xs:enumeration value="useTramService" />
569     <xs:enumeration value="useUndergroundService" />
570     <xs:enumeration value="waitForEscortVehicle" />
571     <xs:enumeration value="other" />
572     </xs:restriction>

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569 </xs:simpleType>
570 <xs:complexType name="GeneralNetworkManagement">
571   <xs:complexContent>
572     <xs:extension base="D2LogicalModel:NetworkManagement">
573       <xs:sequence>
574         <xs:element name="generalNetworkManagementType" type=
575 ↪ "D2LogicalModel:GeneralNetworkManagementTypeEnum" minOccurs="1" maxOccurs="1" />
576         <xs:element name="trafficManuallyDirectedBy" type=
577 ↪ "D2LogicalModel:PersonCategoryEnum" minOccurs="0" maxOccurs="1" />
578         <xs:element name="generalNetworkManagementExtension" type="D2LogicalModel:_
579 ↪ ExtensionType" minOccurs="0" />
580       </xs:sequence>
581     </xs:extension>
582   </xs:complexContent>
583 </xs:complexType>
584 <xs:simpleType name="GeneralNetworkManagementTypeEnum">
585   <xs:restriction base="xs:string">
586     <xs:enumeration value="bridgeSwingInOperation" />
587     <xs:enumeration value="convoyService" />
588     <xs:enumeration value="obstacleSignalling" />
589     <xs:enumeration value="rampMeteringInOperation" />
590     <xs:enumeration value="temporaryTrafficLights" />
591     <xs:enumeration value="tollGatesOpen" />
592     <xs:enumeration value="trafficBeingManuallyDirected" />
593     <xs:enumeration value="trafficHeld" />
594     <xs:enumeration value="other" />
595   </xs:restriction>
596 </xs:simpleType>
597 <xs:simpleType name="GeometryTypeEnum">
598   <xs:restriction base="xs:string" />
599 </xs:simpleType>
600 <xs:complexType name="GlobalNetworkLinear">
601   <xs:sequence>
602     <xs:element name="networkVersion" type="D2LogicalModel:MultilingualString"
603 ↪ minOccurs="1" maxOccurs="1" />
604     <xs:element name="linearGeometryType" type="D2LogicalModel:GeometryTypeEnum"
605 ↪ minOccurs="1" maxOccurs="1" />
606     <xs:element name="startPoint" type="D2LogicalModel:PointByMultiCoordinates"
607 ↪ minOccurs="0" />
608     <xs:element name="endPoint" type="D2LogicalModel:PointByMultiCoordinates"
609 ↪ minOccurs="0" />
610     <xs:element name="linearWithinLinearGNElement" type=
611 ↪ "D2LogicalModel:LinearWithinLinearGNElement" maxOccurs="unbounded" />
612   </xs:sequence>
613 </xs:complexType>
614 <xs:complexType name="GrossWeightCharacteristic">
615   <xs:sequence>
616     <xs:element name="comparisonOperator" type=
617 ↪ "D2LogicalModel:ComparisonOperatorEnum" minOccurs="1" maxOccurs="1" />
618     <xs:element name="grossVehicleWeight" type="D2LogicalModel:Tonnes" minOccurs="1
619 ↪ " maxOccurs="1" />
620     <xs:element name="grossWeightCharacteristicExtension" type="D2LogicalModel:_
621 ↪ ExtensionType" minOccurs="0" />
622   </xs:sequence>
623 </xs:complexType>
624 <xs:complexType name="GroupOfLocations" abstract="true">

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614     <xs:sequence>
615       <xs:element name="groupOfLocationsExtension" type="D2LogicalModel:_ExtensionType
↪ " minOccurs="0" />
616     </xs:sequence>
617   </xs:complexType>
618   <xs:complexType name="HeaderInformation">
619     <xs:sequence>
620       <xs:element name="confidentiality" type="D2LogicalModel:ConfidentialityValueEnum
↪ " minOccurs="1" maxOccurs="1" />
621       <xs:element name="informationStatus" type="D2LogicalModel:InformationStatusEnum
↪ " minOccurs="1" maxOccurs="1" />
622       <xs:element name="urgency" type="D2LogicalModel:UrgencyEnum" minOccurs="0"
↪ maxOccurs="1" />
623       <xs:element name="headerInformationExtension" type="D2LogicalModel:_
↪ ExtensionType" minOccurs="0" />
624     </xs:sequence>
625   </xs:complexType>
626   <xs:complexType name="HeaviestAxleWeightCharacteristic">
627     <xs:sequence>
628       <xs:element name="comparisonOperator" type=
↪ "D2LogicalModel:ComparisonOperatorEnum" minOccurs="1" maxOccurs="1" />
629       <xs:element name="heaviestAxleWeight" type="D2LogicalModel:Tonnes" minOccurs="1
↪ " maxOccurs="1" />
630       <xs:element name="heaviestAxleWeightCharacteristicExtension" type=
↪ "D2LogicalModel:_ExtensionType" minOccurs="0" />
631     </xs:sequence>
632   </xs:complexType>
633   <xs:complexType name="HeightCharacteristic">
634     <xs:sequence>
635       <xs:element name="comparisonOperator" type=
↪ "D2LogicalModel:ComparisonOperatorEnum" minOccurs="1" maxOccurs="1" />
636       <xs:element name="vehicleHeight" type="D2LogicalModel:MetresAsFloat" minOccurs=
↪ "1" maxOccurs="1" />
637       <xs:element name="heightCharacteristicExtension" type="D2LogicalModel:_
↪ ExtensionType" minOccurs="0" />
638     </xs:sequence>
639   </xs:complexType>
640   <xs:simpleType name="HeightGradeEnum">
641     <xs:restriction base="xs:string">
642       <xs:enumeration value="aboveGrade" />
643       <xs:enumeration value="atGrade" />
644       <xs:enumeration value="belowGrade" />
645     </xs:restriction>
646   </xs:simpleType>
647   <xs:complexType name="Impact">
648     <xs:sequence>
649       <xs:element name="capacityRemaining" type="D2LogicalModel:Percentage" minOccurs=
↪ "0" maxOccurs="1" />
650       <xs:element name="numberOfLanesRestricted" type=
↪ "D2LogicalModel:NonNegativeInteger" minOccurs="0" maxOccurs="1" />
651       <xs:element name="numberOfOperationalLanes" type=
↪ "D2LogicalModel:NonNegativeInteger" minOccurs="0" maxOccurs="1" />
652       <xs:element name="originalNumberOfLanes" type="D2LogicalModel:NonNegativeInteger
↪ " minOccurs="0" maxOccurs="1" />
653       <xs:element name="trafficConstrictionType" type=
↪ "D2LogicalModel:TrafficConstrictionTypeEnum" minOccurs="0" maxOccurs="1" />

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654     <xs:element name="delays" type="D2LogicalModel:Delays" minOccurs="0" />
655     <xs:element name="impactExtension" type="D2LogicalModel:_ExtensionType"
↪minOccurs="0" />
656   </xs:sequence>
657 </xs:complexType>
658 <xs:simpleType name="InformationStatusEnum">
659   <xs:restriction base="xs:string">
660     <xs:enumeration value="real" />
661     <xs:enumeration value="securityExercise" />
662     <xs:enumeration value="technicalExercise" />
663     <xs:enumeration value="test" />
664   </xs:restriction>
665 </xs:simpleType>
666 <xs:simpleType name="Integer">
667   <xs:restriction base="xs:integer" />
668 </xs:simpleType>
669 <xs:complexType name="InternationalIdentifier">
670   <xs:sequence>
671     <xs:element name="country" type="D2LogicalModel:CountryEnum" minOccurs="1"
↪maxOccurs="1" />
672     <xs:element name="nationalIdentifier" type="D2LogicalModel:String" minOccurs="1"
↪" maxOccurs="1" />
673     <xs:element name="internationalIdentifierExtension" type="D2LogicalModel:_
↪ExtensionType" minOccurs="0" />
674   </xs:sequence>
675 </xs:complexType>
676 <xs:simpleType name="KilometresPerHour">
677   <xs:restriction base="D2LogicalModel:Float" />
678 </xs:simpleType>
679 <xs:simpleType name="LaneEnum">
680   <xs:restriction base="xs:string">
681     <xs:enumeration value="allLanesCompleteCarriageway" />
682     <xs:enumeration value="busLane" />
683     <xs:enumeration value="busStop" />
684     <xs:enumeration value="carPoolLane" />
685     <xs:enumeration value="centralReservation" />
686     <xs:enumeration value="crawlerLane" />
687     <xs:enumeration value="emergencyLane" />
688     <xs:enumeration value="escapeLane" />
689     <xs:enumeration value="expressLane" />
690     <xs:enumeration value="hardShoulder" />
691     <xs:enumeration value="heavyVehicleLane" />
692     <xs:enumeration value="lane1" />
693     <xs:enumeration value="lane2" />
694     <xs:enumeration value="lane3" />
695     <xs:enumeration value="lane4" />
696     <xs:enumeration value="lane5" />
697     <xs:enumeration value="lane6" />
698     <xs:enumeration value="lane7" />
699     <xs:enumeration value="lane8" />
700     <xs:enumeration value="lane9" />
701     <xs:enumeration value="layBy" />
702     <xs:enumeration value="leftHandTurningLane" />
703     <xs:enumeration value="leftLane" />
704     <xs:enumeration value="localTrafficLane" />
705     <xs:enumeration value="middleLane" />

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706     <xs:enumeration value="opposingLanes" />
707     <xs:enumeration value="overtakingLane" />
708     <xs:enumeration value="rightHandTurningLane" />
709     <xs:enumeration value="rightLane" />
710     <xs:enumeration value="rushHourLane" />
711     <xs:enumeration value="setDownArea" />
712     <xs:enumeration value="slowVehicleLane" />
713     <xs:enumeration value="throughTrafficLane" />
714     <xs:enumeration value="tidalFlowLane" />
715     <xs:enumeration value="turningLane" />
716     <xs:enumeration value="verge" />
717   </xs:restriction>
718 </xs:simpleType>
719 <xs:simpleType name="Language">
720   <xs:restriction base="xs:language" />
721 </xs:simpleType>
722 <xs:complexType name="LengthCharacteristic">
723   <xs:sequence>
724     <xs:element name="comparisonOperator" type=
725 ↪ "D2LogicalModel:ComparisonOperatorEnum" minOccurs="1" maxOccurs="1" />
726     <xs:element name="vehicleLength" type="D2LogicalModel:MetresAsFloat" minOccurs=
727 ↪ "1" maxOccurs="1" />
728     <xs:element name="lengthCharacteristicExtension" type="D2LogicalModel:_
729 ↪ ExtensionType" minOccurs="0" />
730   </xs:sequence>
731 </xs:complexType>
732 <xs:complexType name="Linear">
733   <xs:complexContent>
734     <xs:extension base="D2LogicalModel:NetworkLocation">
735       <xs:sequence>
736         <xs:element name="alertCLinear" type="D2LogicalModel:AlertCLinear"
737 ↪ minOccurs="0" />
738         <xs:element name="linearWithinLinearElement" type=
739 ↪ "D2LogicalModel:LinearWithinLinearElement" minOccurs="0" />
740         <xs:element name="globalNetworkLinear" type=
741 ↪ "D2LogicalModel:GlobalNetworkLinear" minOccurs="0" />
742         <xs:element name="linearExtension" type="D2LogicalModel:_LinearExtensionType"
743 ↪ minOccurs="0" />
744       </xs:sequence>
745     </xs:extension>
746   </xs:complexContent>
747 </xs:complexType>
748 <xs:complexType name="LinearElement">
749   <xs:sequence>
750     <xs:element name="roadName" type="D2LogicalModel:MultilingualString" minOccurs=
751 ↪ "0" maxOccurs="1" />
752     <xs:element name="roadNumber" type="D2LogicalModel:String" minOccurs="0"
753 ↪ maxOccurs="1" />
754     <xs:element name="linearElementReferenceModel" type="D2LogicalModel:String"
755 ↪ minOccurs="0" maxOccurs="1" />
756     <xs:element name="linearElementReferenceModelVersion" type=
757 ↪ "D2LogicalModel:String" minOccurs="0" maxOccurs="1" />
758     <xs:element name="linearElementNature" type=
759 ↪ "D2LogicalModel:LinearElementNatureEnum" minOccurs="0" maxOccurs="1" />
760     <xs:element name="linearElementExtension" type="D2LogicalModel:_ExtensionType"
761 ↪ minOccurs="0" />

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749     </xs:sequence>
750 </xs:complexType>
751 <xs:complexType name="LinearElementByCode">
752   <xs:complexContent>
753     <xs:extension base="D2LogicalModel:LinearElement">
754       <xs:sequence>
755         <xs:element name="linearElementIdentifier" type="D2LogicalModel:String"
756 ↪ minOccurs="1" maxOccurs="1" />
757         <xs:element name="linearElementByCodeExtension" type="D2LogicalModel:_
758 ↪ ExtensionType" minOccurs="0" />
759       </xs:sequence>
760     </xs:extension>
761   </xs:complexContent>
762 </xs:complexType>
763 <xs:complexType name="LinearElementByPoints">
764   <xs:complexContent>
765     <xs:extension base="D2LogicalModel:LinearElement">
766       <xs:sequence>
767         <xs:element name="startPointOfLinearElement" type="D2LogicalModel:Referent"
768 ↪ />
769         <xs:element name="intermediatePointOnLinearElement" type="D2LogicalModel:_
770 ↪ IntermediatePointOnLinearElement" minOccurs="0" maxOccurs="unbounded" />
771         <xs:element name="endPointOfLinearElement" type="D2LogicalModel:Referent" />
772         <xs:element name="linearElementByPointsExtension" type="D2LogicalModel:_
773 ↪ ExtensionType" minOccurs="0" />
774       </xs:sequence>
775     </xs:extension>
776   </xs:complexContent>
777 </xs:complexType>
778 <xs:simpleType name="LinearElementNatureEnum">
779   <xs:restriction base="xs:string">
780     <xs:enumeration value="road" />
781     <xs:enumeration value="roadSection" />
782     <xs:enumeration value="slipRoad" />
783     <xs:enumeration value="other" />
784   </xs:restriction>
785 </xs:simpleType>
786 <xs:simpleType name="LinearReferencingDirectionEnum">
787   <xs:restriction base="xs:string">
788     <xs:enumeration value="both" />
789     <xs:enumeration value="opposite" />
790     <xs:enumeration value="aligned" />
791     <xs:enumeration value="unknown" />
792   </xs:restriction>
793 </xs:simpleType>
794 <xs:complexType name="LinearWithinLinearElement">
795   <xs:sequence>
796     <xs:element name="administrativeAreaOfLinearSection" type=
797 ↪ "D2LogicalModel:MultilingualString" minOccurs="0" maxOccurs="1" />
798     <xs:element name="directionBoundOnLinearSection" type=
799 ↪ "D2LogicalModel:DirectionEnum" minOccurs="0" maxOccurs="1" />
800     <xs:element name="directionRelativeOnLinearSection" type=
801 ↪ "D2LogicalModel:LinearReferencingDirectionEnum" minOccurs="0" maxOccurs="1" />
802     <xs:element name="heightGradeOfLinearSection" type=
803 ↪ "D2LogicalModel:HeightGradeEnum" minOccurs="0" maxOccurs="1" />
804     <xs:element name="linearElement" type="D2LogicalModel:LinearElement" />

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796     <xs:element name="fromPoint" type="D2LogicalModel:DistanceAlongLinearElement" />
797     <xs:element name="toPoint" type="D2LogicalModel:DistanceAlongLinearElement" />
798     <xs:element name="linearWithinLinearElementExtension" type="D2LogicalModel:_
↪ExtensionType" minOccurs="0" />
799   </xs:sequence>
800 </xs:complexType>
801 <xs:complexType name="LinearWithinLinearGNElement">
802   <xs:sequence>
803     <xs:element name="sectionId" type="D2LogicalModel:NonNegativeInteger" minOccurs=
↪"0" maxOccurs="1" />
804     <xs:element name="directionRelativeOnLinearSection" type=
↪"D2LogicalModel:LinearReferencingDirectionEnum" minOccurs="1" maxOccurs="1" />
805     <xs:element name="orderOfSection" type="D2LogicalModel:NonNegativeInteger"
↪minOccurs="0" maxOccurs="1" />
806     <xs:element name="fromPoint" type="D2LogicalModel:DistanceAlongLinearElement" />
807     <xs:element name="toPoint" type="D2LogicalModel:DistanceAlongLinearElement" />
808     <xs:element name="linearWithinLinearGNElementExtension" type="D2LogicalModel:_
↪ExtensionType" minOccurs="0" />
809   </xs:sequence>
810 </xs:complexType>
811 <xs:simpleType name="LoadTypeEnum">
812   <xs:restriction base="xs:string">
813     <xs:enumeration value="abnormalLoad" />
814     <xs:enumeration value="ammunition" />
815     <xs:enumeration value="chemicals" />
816     <xs:enumeration value="combustibleMaterials" />
817     <xs:enumeration value="corrosiveMaterials" />
818     <xs:enumeration value="debris" />
819     <xs:enumeration value="empty" />
820     <xs:enumeration value="explosiveMaterials" />
821     <xs:enumeration value="extraHighLoad" />
822     <xs:enumeration value="extraLongLoad" />
823     <xs:enumeration value="extraWideLoad" />
824     <xs:enumeration value="fuel" />
825     <xs:enumeration value="glass" />
826     <xs:enumeration value="goods" />
827     <xs:enumeration value="hazardousMaterials" />
828     <xs:enumeration value="liquid" />
829     <xs:enumeration value="livestock" />
830     <xs:enumeration value="materials" />
831     <xs:enumeration value="materialsDangerousForPeople" />
832     <xs:enumeration value="materialsDangerousForTheEnvironment" />
833     <xs:enumeration value="materialsDangerousForWater" />
834     <xs:enumeration value="oil" />
835     <xs:enumeration value="ordinary" />
836     <xs:enumeration value="perishableProducts" />
837     <xs:enumeration value="petrol" />
838     <xs:enumeration value="pharmaceuticalMaterials" />
839     <xs:enumeration value="radioactiveMaterials" />
840     <xs:enumeration value="refuse" />
841     <xs:enumeration value="toxicMaterials" />
842     <xs:enumeration value="vehicles" />
843     <xs:enumeration value="other" />
844   </xs:restriction>
845 </xs:simpleType>
846 <xs:complexType name="Location" abstract="true">

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847 <xs:complexContent>
848   <xs:extension base="D2LogicalModel:GroupOfLocations">
849     <xs:sequence>
850       <xs:element name="locationExtension" type="D2LogicalModel:_ExtensionType"
↪ minOccurs="0" />
851     </xs:sequence>
852   </xs:extension>
853 </xs:complexContent>
854 </xs:complexType>
855 <xs:simpleType name="LocationDescriptorEnum">
856   <xs:restriction base="xs:string">
857     <xs:enumeration value="aroundABendInRoad" />
858     <xs:enumeration value="atMotorwayInterchange" />
859     <xs:enumeration value="atRestArea" />
860     <xs:enumeration value="atServiceArea" />
861     <xs:enumeration value="atTollPlaza" />
862     <xs:enumeration value="atTunnelEntryOrExit" />
863     <xs:enumeration value="inbound" />
864     <xs:enumeration value="inGallery" />
865     <xs:enumeration value="inTheCentre" />
866     <xs:enumeration value="inTheOppositeDirection" />
867     <xs:enumeration value="inTunnel" />
868     <xs:enumeration value="onBorder" />
869     <xs:enumeration value="onBridge" />
870     <xs:enumeration value="onConnector" />
871     <xs:enumeration value="onElevatedSection" />
872     <xs:enumeration value="onFlyover" />
873     <xs:enumeration value="onIceRoad" />
874     <xs:enumeration value="onLevelCrossing" />
875     <xs:enumeration value="onLinkRoad" />
876     <xs:enumeration value="onPass" />
877     <xs:enumeration value="onRoundabout" />
878     <xs:enumeration value="onTheLeft" />
879     <xs:enumeration value="onTheRight" />
880     <xs:enumeration value="onTheRoadway" />
881     <xs:enumeration value="onUndergroundSection" />
882     <xs:enumeration value="onUnderpass" />
883     <xs:enumeration value="outbound" />
884     <xs:enumeration value="overCrestOfHill" />
885     <xs:enumeration value="withinJunction" />
886   </xs:restriction>
887 </xs:simpleType>
888 <xs:simpleType name="MaintenanceVehicleActionsEnum">
889   <xs:restriction base="xs:string">
890     <xs:enumeration value="maintenanceVehiclesMergingIntoTrafficFlow" />
891     <xs:enumeration value="saltAndGritSpreading" />
892     <xs:enumeration value="slowMoving" />
893     <xs:enumeration value="snowClearing" />
894     <xs:enumeration value="stoppingToServiceEquipments" />
895   </xs:restriction>
896 </xs:simpleType>
897 <xs:complexType name="MaintenanceVehicles">
898   <xs:sequence>
899     <xs:element name="numberOfMaintenanceVehicles" type=
↪ "D2LogicalModel:NonNegativeInteger" minOccurs="0" maxOccurs="1" />
900     <xs:element name="maintenanceVehicleActions" type=

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↪ "D2LogicalModel:MaintenanceVehicleActionsEnum" minOccurs="0" maxOccurs="unbounded" /
↪ >
901   <xs:element name="maintenanceVehiclesExtension" type="D2LogicalModel:_
↪ ExtensionType" minOccurs="0" />
902   </xs:sequence>
903   </xs:complexType>
904   <xs:complexType name="MaintenanceWorks">
905     <xs:complexContent>
906       <xs:extension base="D2LogicalModel:Roadworks">
907         <xs:sequence>
908           <xs:element name="roadMaintenanceType" type=
↪ "D2LogicalModel:RoadMaintenanceTypeEnum" minOccurs="1" maxOccurs="unbounded" />
909           <xs:element name="maintenanceWorksExtension" type="D2LogicalModel:_
↪ ExtensionType" minOccurs="0" />
910         </xs:sequence>
911       </xs:extension>
912     </xs:complexContent>
913   </xs:complexType>
914   <xs:complexType name="MeasurementEquipmentFault">
915     <xs:complexContent>
916       <xs:extension base="D2LogicalModel:Fault">
917         <xs:sequence>
918           <xs:element name="measurementEquipmentFault" type=
↪ "D2LogicalModel:MeasurementEquipmentFaultEnum" minOccurs="1" maxOccurs="1" />
919           <xs:element name="measurementEquipmentFaultExtension" type="D2LogicalModel:_
↪ ExtensionType" minOccurs="0" />
920         </xs:sequence>
921       </xs:extension>
922     </xs:complexContent>
923   </xs:complexType>
924   <xs:simpleType name="MeasurementEquipmentFaultEnum">
925     <xs:restriction base="xs:string">
926       <xs:enumeration value="intermittentDataValues" />
927       <xs:enumeration value="noDataValuesAvailable" />
928       <xs:enumeration value="spuriousUnreliableDataValues" />
929       <xs:enumeration value="unspecifiedOrUnknownFault" />
930       <xs:enumeration value="other" />
931     </xs:restriction>
932   </xs:simpleType>
933   <xs:simpleType name="MetresAsFloat">
934     <xs:restriction base="D2LogicalModel:Float" />
935   </xs:simpleType>
936   <xs:simpleType name="MetresAsNonNegativeInteger">
937     <xs:restriction base="D2LogicalModel:NonNegativeInteger" />
938   </xs:simpleType>
939   <xs:complexType name="Mobility">
940     <xs:sequence>
941       <xs:element name="mobilityType" type="D2LogicalModel:MobilityEnum" minOccurs="1
↪ " maxOccurs="1" />
942       <xs:element name="mobilityExtension" type="D2LogicalModel:_ExtensionType"
↪ minOccurs="0" />
943     </xs:sequence>
944   </xs:complexType>
945   <xs:simpleType name="MobilityEnum">
946     <xs:restriction base="xs:string">
947       <xs:enumeration value="mobile" />

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948     <xs:enumeration value="stationary" />
949     <xs:enumeration value="unknown" />
950   </xs:restriction>
951 </xs:simpleType>
952 <xs:complexType name="MultilingualString">
953   <xs:sequence>
954     <xs:element name="values">
955       <xs:complexType>
956         <xs:sequence>
957           <xs:element name="value" type="D2LogicalModel:MultilingualStringValue"
958             ↪ maxOccurs="unbounded" />
959         </xs:sequence>
960       </xs:complexType>
961     </xs:element>
962   </xs:sequence>
963 </xs:complexType>
964 <xs:complexType name="MultilingualStringValue">
965   <xs:simpleContent>
966     <xs:extension base="D2LogicalModel:MultilingualStringValue" type="D2LogicalModel:MultilingualStringValue"
967       ↪ base="D2LogicalModel:MultilingualStringValue" />
968     <xs:attribute name="lang" type="xs:language" />
969   </xs:simpleContent>
970 </xs:complexType>
971 <xs:simpleType name="MultilingualStringValue" type="D2LogicalModel:MultilingualStringValue" />
972 <xs:restriction base="xs:string">
973   <xs:maxLength value="1024" />
974 </xs:restriction>
975 </xs:simpleType>
976 <xs:complexType name="NetworkLocation" abstract="true">
977   <xs:complexContent>
978     <xs:extension base="D2LogicalModel:Location">
979       <xs:sequence>
980         <xs:element name="supplementaryPositionalDescription" type="D2LogicalModel:SupplementaryPositionalDescription"
981           ↪ minOccurs="0" />
982         <xs:element name="destination" type="D2LogicalModel:Destination" minOccurs="0" />
983         <xs:element name="networkLocationExtension" type="D2LogicalModel:NetworkLocationExtension"
984           ↪ minOccurs="0" />
985       </xs:sequence>
986     </xs:extension>
987   </xs:complexContent>
988 </xs:complexType>
989 <xs:complexType name="NetworkManagement" abstract="true">
990   <xs:complexContent>
991     <xs:extension base="D2LogicalModel:OperatorAction">
992       <xs:sequence>
993         <xs:element name="complianceOption" type="D2LogicalModel:ComplianceOptionEnum"
994           ↪ minOccurs="1" maxOccurs="1" />
995         <xs:element name="applicableForTrafficDirection" type="D2LogicalModel:DirectionEnum"
996           ↪ minOccurs="0" maxOccurs="unbounded" />
997         <xs:element name="applicableForTrafficType" type="D2LogicalModel:TrafficTypeEnum"
998           ↪ minOccurs="0" maxOccurs="unbounded" />
999         <xs:element name="placesAtWhichApplicable" type="D2LogicalModel:PlacesEnum"
1000           ↪ minOccurs="0" maxOccurs="unbounded" />
1001         <xs:element name="forVehiclesWithCharacteristicsOf" type="D2LogicalModel:VehicleCharacteristics"
1002           ↪ minOccurs="0" maxOccurs="unbounded" />

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995     <xs:element name="networkManagementExtension" type="D2LogicalModel:_
↪ExtensionType" minOccurs="0" />
996     </xs:sequence>
997   </xs:extension>
998   </xs:complexContent>
999 </xs:complexType>
1000 <xs:complexType name="NonManagedCause">

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1001   <xs:complexContent>
1002     <xs:extension base="D2LogicalModel:Cause">
1003       <xs:sequence>
1004         <xs:element name="causeType" type="D2LogicalModel:CauseTypeEnum" minOccurs=
↪"0" maxOccurs="1" />
1005         <xs:element name="nonManagedCauseExtension" type="D2LogicalModel:_
↪ExtensionType" minOccurs="0" />
1006       </xs:sequence>
1007     </xs:extension>
1008   </xs:complexContent>
1009 </xs:complexType>
1010 <xs:simpleType name="NonNegativeInteger">
1011   <xs:restriction base="xs:nonNegativeInteger" />
1012 </xs:simpleType>
1013 <xs:complexType name="NumberOfAxlesCharacteristic">
1014   <xs:sequence>
1015     <xs:element name="comparisonOperator" type=
↪"D2LogicalModel:ComparisonOperatorEnum" minOccurs="1" maxOccurs="1" />
1016     <xs:element name="numberOfAxles" type="D2LogicalModel:NonNegativeInteger"
↪minOccurs="1" maxOccurs="1" />
1017     <xs:element name="numberOfAxlesCharacteristicExtension" type="D2LogicalModel:_
↪ExtensionType" minOccurs="0" />
1018   </xs:sequence>
1019 </xs:complexType>
1020 <xs:complexType name="OccupancyChangeValue">
1021   <xs:complexContent>
1022     <xs:extension base="D2LogicalModel:DataValue">
1023       <xs:sequence>
1024         <xs:element name="occupancyChange" type="D2LogicalModel:Integer" minOccurs=
↪"1" maxOccurs="1" />
1025         <xs:element name="occupancyChangeValueExtension" type="D2LogicalModel:_
↪ExtensionType" minOccurs="0" />
1026       </xs:sequence>
1027     </xs:extension>
1028   </xs:complexContent>
1029 </xs:complexType>
1030 <xs:complexType name="OpenlrBaseLocationReferencePoint" abstract="true">
1031   <xs:sequence>
1032     <xs:element name="openlrCoordinate" type="D2LogicalModel:PointCoordinates" />
1033     <xs:element name="openlrLineAttributes" type=
↪"D2LogicalModel:OpenlrLineAttributes" />
1034     <xs:element name="openlrBaseLocationReferencePointExtension" type=
↪"D2LogicalModel:_ExtensionType" minOccurs="0" />
1035   </xs:sequence>
1036 </xs:complexType>
1037 <xs:complexType name="OpenlrBasePointLocation" abstract="true">
1038   <xs:sequence>
1039     <xs:element name="openlrSideOfRoad" type="D2LogicalModel:OpenlrSideOfRoadEnum"
↪

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1040   <minOccurs="1" maxOccurs="1" />
1041   <xs:element name="openlrOrientation" type="D2LogicalModel:OpenlrOrientationEnum"
1042   <minOccurs="1" maxOccurs="1" />
1043   <xs:element name="openlrPositiveOffset" type=
1044   <"D2LogicalModel:MetresAsNonNegativeInteger" minOccurs="0" maxOccurs="1" />
1045   <xs:element name="openlrLocationReferencePoint" type=
1046   <"D2LogicalModel:OpenlrLocationReferencePoint" />
1047   <xs:element name="openlrLastLocationReferencePoint" type=
1048   <"D2LogicalModel:OpenlrLastLocationReferencePoint" />
1049   <xs:element name="openlrBasePointLocationExtension" type="D2LogicalModel:_
1050   <ExtensionType" minOccurs="0" />
1051   </xs:sequence>
1052   </xs:complexType>
1053   <xs:complexType name="OpenlrExtendedLinear">
1054   <xs:sequence>
1055   <xs:element name="firstDirection" type=
1056   <"D2LogicalModel:OpenlrLineLocationReference" />
1057   <xs:element name="oppositeDirection" type=
1058   <"D2LogicalModel:OpenlrLineLocationReference" minOccurs="0" />
1059   </xs:sequence>
1060   </xs:complexType>
1061   <xs:complexType name="OpenlrExtendedPoint">
1062   <xs:sequence>
1063   <xs:element name="openlrPointLocationReference" type=
1064   <"D2LogicalModel:OpenlrPointLocationReference" />
1065   </xs:sequence>
1066   </xs:complexType>
1067   <xs:simpleType name="OpenlrFormOfWayEnum">
1068   <xs:restriction base="xs:string">
1069   <xs:enumeration value="undefined" />
1070   <xs:enumeration value="motorway" />
1071   <xs:enumeration value="multipleCarriageway" />
1072   <xs:enumeration value="singleCarriageway" />
1073   <xs:enumeration value="roundabout" />
1074   <xs:enumeration value="slipRoad" />
1075   <xs:enumeration value="trafficSquare" />
1076   <xs:enumeration value="other" />
1077   </xs:restriction>
1078   </xs:simpleType>
1079   <xs:simpleType name="OpenlrFunctionalRoadClassEnum">
1080   <xs:restriction base="xs:string">
1081   <xs:enumeration value="FRC0" />
1082   <xs:enumeration value="FRC1" />
1083   <xs:enumeration value="FRC2" />
1084   <xs:enumeration value="FRC3" />
1085   <xs:enumeration value="FRC4" />
1086   <xs:enumeration value="FRC5" />
1087   <xs:enumeration value="FRC6" />
1088   <xs:enumeration value="FRC7" />
1089   </xs:restriction>
1090   </xs:simpleType>
1091   <xs:complexType name="OpenlrGeoCoordinate">
1092   <xs:sequence>
1093   <xs:element name="openlrCoordinate" type="D2LogicalModel:PointCoordinates" />
1094   <xs:element name="openlrGeoCoordinateExtension" type="D2LogicalModel:_
1095   <ExtensionType" minOccurs="0" />

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1086     </xs:sequence>
1087 </xs:complexType>
1088 <xs:complexType name="OpenlrLastLocationReferencePoint">
1089     <xs:complexContent>
1090         <xs:extension base="D2LogicalModel:OpenlrBaseLocationReferencePoint">
1091             <xs:sequence>
1092                 <xs:element name="openlrLastLocationReferencePointExtension" type=
1093 ↪ "D2LogicalModel:_ExtensionType" minOccurs="0" />
1094             </xs:sequence>
1095         </xs:extension>
1096     </xs:complexContent>
1097 </xs:complexType>
1098 <xs:complexType name="OpenlrLineAttributes">
1099     <xs:sequence>
1100         <xs:element name="openlrFunctionalRoadClass" type=
1101 ↪ "D2LogicalModel:OpenlrFunctionalRoadClassEnum" minOccurs="1" maxOccurs="1" />
1102         <xs:element name="openlrFormOfWay" type="D2LogicalModel:OpenlrFormOfWayEnum"
1103 ↪ minOccurs="1" maxOccurs="1" />
1104         <xs:element name="openlrBearing" type="D2LogicalModel:AngleInDegrees" minOccurs=
1105 ↪ "1" maxOccurs="1" />
1106         <xs:element name="openlrLineAttributesExtension" type="D2LogicalModel:_
1107 ↪ ExtensionType" minOccurs="0" />
1108     </xs:sequence>
1109 </xs:complexType>
1110 <xs:complexType name="OpenlrLineLocationReference">
1111     <xs:sequence>
1112         <xs:element name="openlrLocationReferencePoint" type=
1113 ↪ "D2LogicalModel:OpenlrLocationReferencePoint" maxOccurs="unbounded" />
1114         <xs:element name="openlrLastLocationReferencePoint" type=
1115 ↪ "D2LogicalModel:OpenlrLastLocationReferencePoint" />
1116         <xs:element name="openlrOffsets" type="D2LogicalModel:OpenlrOffsets" minOccurs=
1117 ↪ "0" />
1118         <xs:element name="openlrLineLocationReferenceExtension" type="D2LogicalModel:_
1119 ↪ ExtensionType" minOccurs="0" />
1120     </xs:sequence>
1121 </xs:complexType>
1122 <xs:complexType name="OpenlrLocationReferencePoint">
1123     <xs:complexContent>
1124         <xs:extension base="D2LogicalModel:OpenlrBaseLocationReferencePoint">
1125             <xs:sequence>
1126                 <xs:element name="openlrPathAttributes" type=
1127 ↪ "D2LogicalModel:OpenlrPathAttributes" />
1128                 <xs:element name="openlrLocationReferencePointExtension" type=
1129 ↪ "D2LogicalModel:_ExtensionType" minOccurs="0" />
1130             </xs:sequence>
1131         </xs:extension>
1132     </xs:complexContent>
1133 </xs:complexType>
1134 <xs:complexType name="OpenlrOffsets">
1135     <xs:sequence>
1136         <xs:element name="openlrPositiveOffset" type=
1137 ↪ "D2LogicalModel:MetresAsNonNegativeInteger" minOccurs="0" maxOccurs="1" />
1138         <xs:element name="openlrNegativeOffset" type=
1139 ↪ "D2LogicalModel:MetresAsNonNegativeInteger" minOccurs="0" maxOccurs="1" />
1140         <xs:element name="openlrOffsetsExtension" type="D2LogicalModel:_ExtensionType"
1141 ↪ minOccurs="0" />

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1128     </xs:sequence>
1129 </xs:complexType>
1130 <xs:simpleType name="OpenlrOrientationEnum">
1131     <xs:restriction base="xs:string">
1132         <xs:enumeration value="noOrientationOrUnknown" />
1133         <xs:enumeration value="withLineDirection" />
1134         <xs:enumeration value="againstLineDirection" />
1135         <xs:enumeration value="both" />
1136     </xs:restriction>
1137 </xs:simpleType>
1138 <xs:complexType name="OpenlrPathAttributes">
1139     <xs:sequence>
1140         <xs:element name="openlrLowestFRCToNextLRPoint" type=
1141 ↪ "D2LogicalModel:OpenlrFunctionalRoadClassEnum" minOccurs="1" maxOccurs="1" />
1142         <xs:element name="openlrDistanceToNextLRPoint" type=
1143 ↪ "D2LogicalModel:NonNegativeInteger" minOccurs="1" maxOccurs="1" />
1144         <xs:element name="openlrPathAttributesExtension" type="D2LogicalModel:_
1145 ↪ ExtensionType" minOccurs="0" />
1146     </xs:sequence>
1147 </xs:complexType>
1148 <xs:complexType name="OpenlrPointAlongLine">
1149     <xs:complexContent>
1150         <xs:extension base="D2LogicalModel:OpenlrBasePointLocation">
1151             <xs:sequence>
1152                 <xs:element name="openlrPointAlongLineExtension" type="D2LogicalModel:_
1153 ↪ ExtensionType" minOccurs="0" />
1154             </xs:sequence>
1155         </xs:extension>
1156     </xs:complexContent>
1157 </xs:complexType>
1158 <xs:complexType name="OpenlrPointLocationReference">
1159     <xs:sequence>
1160         <xs:element name="openlrGeoCoordinate" type="D2LogicalModel:OpenlrGeoCoordinate
1161 ↪ " minOccurs="0" />
1162         <xs:element name="openlrPoiWithAccessPoint" type=
1163 ↪ "D2LogicalModel:OpenlrPoiWithAccessPoint" minOccurs="0" />
1164         <xs:element name="openlrPointAlongLine" type=
1165 ↪ "D2LogicalModel:OpenlrPointAlongLine" minOccurs="0" />
1166         <xs:element name="openlrPointLocationReferenceExtension" type="D2LogicalModel:_
1167 ↪ ExtensionType" minOccurs="0" />
1168     </xs:sequence>
1169 </xs:complexType>
1170 <xs:complexType name="OpenlrPoiWithAccessPoint">
1171     <xs:complexContent>
1172         <xs:extension base="D2LogicalModel:OpenlrBasePointLocation">
1173             <xs:sequence>
1174                 <xs:element name="openlrCoordinate" type="D2LogicalModel:PointCoordinates" /
1175 ↪ >
1176                 <xs:element name="openlrPoiWithAccessPointExtension" type="D2LogicalModel:_
1177 ↪ ExtensionType" minOccurs="0" />
1178             </xs:sequence>
1179         </xs:extension>
1180     </xs:complexContent>
1181 </xs:complexType>
1182 <xs:simpleType name="OpenlrSideOfRoadEnum">
1183     <xs:restriction base="xs:string">

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1174     <xs:enumeration value="onRoadOrUnknown" />
1175     <xs:enumeration value="right" />
1176     <xs:enumeration value="left" />
1177     <xs:enumeration value="both" />
1178   </xs:restriction>
1179 </xs:simpleType>
1180 <xs:complexType name="OperatorAction">
1181   <xs:complexContent>
1182     <xs:extension base="D2LogicalModel:SituationRecord">
1183       <xs:sequence>
1184         <xs:element name="operatorActionExtension" type="D2LogicalModel:_
↪ ExtensionType" minOccurs="0" />
1185       </xs:sequence>
1186     </xs:extension>
1187   </xs:complexContent>
1188 </xs:complexType>
1189 <xs:complexType name="OverallPeriod">
1190   <xs:sequence>
1191     <xs:element name="overallStartTime" type="D2LogicalModel:DateTime" minOccurs="1
↪ " maxOccurs="1" />
1192     <xs:element name="overallEndTime" type="D2LogicalModel:DateTime" minOccurs="0"
↪ maxOccurs="1" />
1193     <xs:element name="overallPeriodExtension" type="D2LogicalModel:_ExtensionType"
↪ minOccurs="0" />
1194   </xs:sequence>
1195 </xs:complexType>
1196 <xs:simpleType name="PassengerCarUnitsPerHour">
1197   <xs:restriction base="D2LogicalModel:NonNegativeInteger" />
1198 </xs:simpleType>
1199 <xs:complexType name="PayloadPublication" abstract="true">
1200   <xs:sequence>
1201     <xs:element name="feedType" type="D2LogicalModel:String" minOccurs="0"
↪ maxOccurs="1" />
1202     <xs:element name="publicationTime" type="D2LogicalModel:DateTime" minOccurs="1"
↪ maxOccurs="1" />
1203     <xs:element name="publicationCreator" type=
↪ "D2LogicalModel:InternationalIdentifier" />
1204     <xs:element name="payloadPublicationExtension" type="D2LogicalModel:_
↪ ExtensionType" minOccurs="0" />
1205   </xs:sequence>
1206   <xs:attribute name="lang" type="D2LogicalModel:Language" use="required" />
1207 </xs:complexType>
1208 <xs:complexType name="PcuFlowValue">
1209   <xs:complexContent>
1210     <xs:extension base="D2LogicalModel:DataValue">
1211       <xs:sequence>
1212         <xs:element name="pcuFlowRate" type="D2LogicalModel:PassengerCarUnitsPerHour
↪ " minOccurs="1" maxOccurs="1" />
1213         <xs:element name="pcuFlowValueExtension" type="D2LogicalModel:_ExtensionType
↪ " minOccurs="0" />
1214       </xs:sequence>
1215     </xs:extension>
1216   </xs:complexContent>
1217 </xs:complexType>
1218 <xs:simpleType name="Percentage">
1219   <xs:restriction base="D2LogicalModel:Float" />

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1220 </xs:simpleType>
1221 <xs:complexType name="PercentageDistanceAlongLinearElement">
1222   <xs:complexContent>
1223     <xs:extension base="D2LogicalModel:DistanceAlongLinearElement">
1224       <xs:sequence>
1225         <xs:element name="percentageDistanceAlong" type="D2LogicalModel:Percentage"
1226           ↪ minOccurs="1" maxOccurs="1" />
1227         <xs:element name="percentageDistanceAlongLinearElementExtension" type=
1228           ↪ "D2LogicalModel:_ExtensionType" minOccurs="0" />
1229       </xs:sequence>
1230     </xs:extension>
1231   </xs:complexContent>
1232 </xs:complexType>
1233 <xs:simpleType name="PersonCategoryEnum">
1234   <xs:restriction base="xs:string">
1235     <xs:enumeration value="adult" />
1236     <xs:enumeration value="child" />
1237     <xs:enumeration value="emergencyServicesPerson" />
1238     <xs:enumeration value="fireman" />
1239     <xs:enumeration value="infant" />
1240     <xs:enumeration value="medicalStaff" />
1241     <xs:enumeration value="memberOfThePublic" />
1242     <xs:enumeration value="policeman" />
1243     <xs:enumeration value="politician" />
1244     <xs:enumeration value="publicTransportPassenger" />
1245     <xs:enumeration value="sickPerson" />
1246     <xs:enumeration value="trafficOfficer" />
1247     <xs:enumeration value="trafficWarden" />
1248     <xs:enumeration value="veryImportantPerson" />
1249   </xs:restriction>
1250 </xs:simpleType>
1251 <xs:simpleType name="PlacesEnum">
1252   <xs:restriction base="xs:string">
1253     <xs:enumeration value="aroundBendsInTheRoad" />
1254     <xs:enumeration value="atCustomsPosts" />
1255     <xs:enumeration value="atHighAltitudes" />
1256     <xs:enumeration value="atTollPlazas" />
1257     <xs:enumeration value="inBuiltUpAreas" />
1258     <xs:enumeration value="inContraflowSections" />
1259     <xs:enumeration value="inForestedAreas" />
1260     <xs:enumeration value="inGalleries" />
1261     <xs:enumeration value="inLowLyingAreas" />
1262     <xs:enumeration value="inRoadworksAreas" />
1263     <xs:enumeration value="inRuralAreas" />
1264     <xs:enumeration value="inShadedAreas" />
1265     <xs:enumeration value="inTheCityCentre" />
1266     <xs:enumeration value="inTheInnerCityAreas" />
1267     <xs:enumeration value="inTunnels" />
1268     <xs:enumeration value="onBridges" />
1269     <xs:enumeration value="onDownHillSections" />
1270     <xs:enumeration value="onDualCarriagewaySections" />
1271     <xs:enumeration value="onElevatedSections" />
1272     <xs:enumeration value="onEnteringOrLeavingTunnels" />
1273     <xs:enumeration value="onEnteringTheCountry" />
1274     <xs:enumeration value="onFlyovers" />
1275     <xs:enumeration value="onLeavingTheCountry" />

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1274     <xs:enumeration value="onMotorways" />
1275     <xs:enumeration value="onNonMotorways" />
1276     <xs:enumeration value="onPasses" />
1277     <xs:enumeration value="onRoundabouts" />
1278     <xs:enumeration value="onSingleCarriagewaySections" />
1279     <xs:enumeration value="onSlipRoads" />
1280     <xs:enumeration value="onUndergroundSections" />
1281     <xs:enumeration value="onUnderpasses" />
1282     <xs:enumeration value="onUpHillSections" />
1283     <xs:enumeration value="overTheCrestOfHills" />
1284     <xs:enumeration value="other" />
1285   </xs:restriction>
1286 </xs:simpleType>
1287 <xs:complexType name="Point">
1288   <xs:complexContent>
1289     <xs:extension base="D2LogicalModel:NetworkLocation">
1290       <xs:sequence>
1291         <xs:element name="alertCPoint" type="D2LogicalModel:AlertCPoint" minOccurs=
1292 ↪ "0" />
1293         <xs:element name="pointAlongLinearElement" type=
1294 ↪ "D2LogicalModel:PointAlongLinearElement" minOccurs="0" />
1295         <xs:element name="pointWithinLinearGNElement" type=
1296 ↪ "D2LogicalModel:PointWithinLinearGNElement" minOccurs="0" />
1297         <xs:element name="pointExtension" type="D2LogicalModel:_PointExtensionType"
1298 ↪ minOccurs="0" />
1299       </xs:sequence>
1300     </xs:extension>
1301   </xs:complexContent>
1302 </xs:complexType>
1303 <xs:complexType name="PointAlongLinearElement">
1304   <xs:sequence>
1305     <xs:element name="administrativeAreaOfPoint" type=
1306 ↪ "D2LogicalModel:MultilingualString" minOccurs="0" maxOccurs="1" />
1307     <xs:element name="directionBoundAtPoint" type="D2LogicalModel:DirectionEnum"
1308 ↪ minOccurs="0" maxOccurs="1" />
1309     <xs:element name="directionRelativeAtPoint" type=
1310 ↪ "D2LogicalModel:LinearReferencingDirectionEnum" minOccurs="0" maxOccurs="1" />
1311     <xs:element name="heightGradeOfPoint" type="D2LogicalModel:HeightGradeEnum"
1312 ↪ minOccurs="0" maxOccurs="1" />
1313     <xs:element name="linearElement" type="D2LogicalModel:LinearElement" />
1314     <xs:element name="distanceAlongLinearElement" type=
1315 ↪ "D2LogicalModel:DistanceAlongLinearElement" />
1316     <xs:element name="pointAlongLinearElementExtension" type="D2LogicalModel:_
1317 ↪ ExtensionType" minOccurs="0" />
1318   </xs:sequence>
1319 </xs:complexType>
1320 <xs:complexType name="PointByMultiCoordinates">
1321   <xs:sequence>
1322     <xs:element name="sjtskPointCoordinates" type=
1323 ↪ "D2LogicalModel:SJTSKPointCoordinates" minOccurs="0" />
1324     <xs:element name="pointCoordinates" type="D2LogicalModel:PointCoordinates"
1325 ↪ minOccurs="0" />
1326     <xs:element name="pointByMultiCoordinatesExtension" type="D2LogicalModel:_
1327 ↪ ExtensionType" minOccurs="0" />
1328   </xs:sequence>
1329 </xs:complexType>

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1317 <xs:complexType name="PointCoordinates">
1318   <xs:sequence>
1319     <xs:element name="latitude" type="D2LogicalModel:Float" minOccurs="1" maxOccurs=
1320 ↪ "1" />
1321     <xs:element name="longitude" type="D2LogicalModel:Float" minOccurs="1"
1322 ↪ maxOccurs="1" />
1323     <xs:element name="pointCoordinatesExtension" type="D2LogicalModel:_ExtensionType
1324 ↪ " minOccurs="0" />
1325   </xs:sequence>
1326 </xs:complexType>
1327 <xs:complexType name="PointDestination">
1328   <xs:complexContent>
1329     <xs:extension base="D2LogicalModel:Destination">
1330       <xs:sequence>
1331         <xs:element name="point" type="D2LogicalModel:Point" />
1332         <xs:element name="pointDestinationExtension" type="D2LogicalModel:_
1333 ↪ ExtensionType" minOccurs="0" />
1334       </xs:sequence>
1335     </xs:extension>
1336   </xs:complexContent>
1337 </xs:complexType>
1338 <xs:complexType name="PointWithinLinearGNElement">
1339   <xs:sequence>
1340     <xs:element name="sectionId" type="D2LogicalModel:NonNegativeInteger" minOccurs=
1341 ↪ "0" maxOccurs="1" />
1342     <xs:element name="directionRelativeOnLinearSection" type=
1343 ↪ "D2LogicalModel:LinearReferencingDirectionEnum" minOccurs="1" maxOccurs="1" />
1344     <xs:element name="networkVersion" type="D2LogicalModel:MultilingualString"
1345 ↪ minOccurs="1" maxOccurs="1" />
1346     <xs:element name="distanceAlongLinearElement" type=
1347 ↪ "D2LogicalModel:DistanceAlongLinearElement" />
1348     <xs:element name="pointWithinLinearGNElementExtension" type="D2LogicalModel:_
1349 ↪ ExtensionType" minOccurs="0" />
1350   </xs:sequence>
1351 </xs:complexType>
1352 <xs:simpleType name="ProbabilityOfOccurrenceEnum">
1353   <xs:restriction base="xs:string">
1354     <xs:enumeration value="certain" />
1355     <xs:enumeration value="probable" />
1356     <xs:enumeration value="riskOf" />
1357   </xs:restriction>
1358 </xs:simpleType>
1359 <xs:complexType name="Referent">
1360   <xs:sequence>
1361     <xs:element name="referentIdentifier" type="D2LogicalModel:String" minOccurs="1
1362 ↪ " maxOccurs="1" />
1363     <xs:element name="referentName" type="D2LogicalModel:String" minOccurs="0"
1364 ↪ maxOccurs="1" />
1365     <xs:element name="referentType" type="D2LogicalModel:ReferentTypeEnum"
1366 ↪ minOccurs="1" maxOccurs="1" />
1367     <xs:element name="referentDescription" type="D2LogicalModel:MultilingualString"
1368 ↪ minOccurs="0" maxOccurs="1" />
1369     <xs:element name="pointCoordinates" type="D2LogicalModel:PointCoordinates"
1370 ↪ minOccurs="0" />
1371     <xs:element name="referentExtension" type="D2LogicalModel:_ExtensionType"
1372 ↪ minOccurs="0" />

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1358     </xs:sequence>
1359 </xs:complexType>
1360 <xs:simpleType name="ReferentTypeEnum">
1361     <xs:restriction base="xs:string">
1362         <xs:enumeration value="boundary" />
1363         <xs:enumeration value="intersection" />
1364         <xs:enumeration value="referenceMarker" />
1365         <xs:enumeration value="landmark" />
1366         <xs:enumeration value="roadNode" />
1367     </xs:restriction>
1368 </xs:simpleType>
1369 <xs:complexType name="ReroutingManagement">
1370     <xs:complexContent>
1371         <xs:extension base="D2LogicalModel:NetworkManagement">
1372             <xs:sequence>
1373                 <xs:element name="reroutingManagementType" type=
1374 ↪ "D2LogicalModel:ReroutingManagementTypeEnum" minOccurs="1" maxOccurs="unbounded" />
1375                 <xs:element name="reroutingItineraryDescription" type=
1376 ↪ "D2LogicalModel:MultilingualString" minOccurs="0" maxOccurs="1" />
1377                 <xs:element name="signedRerouting" type="D2LogicalModel:Boolean" minOccurs=
1378 ↪ "0" maxOccurs="1" />
1379                 <xs:element name="entry" type="D2LogicalModel:String" minOccurs="0"
1380 ↪ maxOccurs="1" />
1381                 <xs:element name="exit" type="D2LogicalModel:String" minOccurs="0"
1382 ↪ maxOccurs="1" />
1383                 <xs:element name="roadOrJunctionNumber" type="D2LogicalModel:String"
1384 ↪ minOccurs="0" maxOccurs="1" />
1385                 <xs:element name="reroutingManagementExtension" type="D2LogicalModel:_
1386 ↪ ExtensionType" minOccurs="0" />
1387             </xs:sequence>
1388         </xs:extension>
1389     </xs:complexContent>
1390 </xs:complexType>
1391 <xs:simpleType name="ReroutingManagementTypeEnum">
1392     <xs:restriction base="xs:string">
1393         <xs:enumeration value="doNotFollowDiversionSigns" />
1394         <xs:enumeration value="doNotUseEntry" />
1395         <xs:enumeration value="doNotUseExit" />
1396         <xs:enumeration value="doNotUseIntersectionOrJunction" />
1397         <xs:enumeration value="followDiversionSigns" />
1398         <xs:enumeration value="followLocalDiversion" />
1399         <xs:enumeration value="followSpecialMarkers" />
1400         <xs:enumeration value="useEntry" />
1401         <xs:enumeration value="useExit" />
1402         <xs:enumeration value="useIntersectionOrJunction" />
1403     </xs:restriction>
1404 </xs:simpleType>
1405 <xs:simpleType name="RoadMaintenanceTypeEnum">
1406     <xs:restriction base="xs:string">
1407         <xs:enumeration value="clearanceWork" />
1408         <xs:enumeration value="controlledAvalanche" />
1409         <xs:enumeration value="installationWork" />
1410         <xs:enumeration value="grassCuttingWork" />
1411         <xs:enumeration value="litterClearance" />
1412         <xs:enumeration value="maintenanceWork" />
1413         <xs:enumeration value="overheadWorks" />

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1407     <xs:enumeration value="repairWork" />
1408     <xs:enumeration value="resurfacingWork" />
1409     <xs:enumeration value="roadMarkingWork" />
1410     <xs:enumeration value="roadsideWork" />
1411     <xs:enumeration value="roadworksClearance" />
1412     <xs:enumeration value="roadworks" />
1413     <xs:enumeration value="rockFallPreventativeMaintenance" />
1414     <xs:enumeration value="saltingInProgress" />
1415     <xs:enumeration value="snowploughsInUse" />
1416     <xs:enumeration value="sweepingOfRoad" />
1417     <xs:enumeration value="treeAndVegetationCuttingWork" />
1418     <xs:enumeration value="other" />
1419   </xs:restriction>
1420 </xs:simpleType>
1421 <xs:complexType name="RoadOrCarriagewayOrLaneManagement">
1422   <xs:complexContent>
1423     <xs:extension base="D2LogicalModel:NetworkManagement">
1424       <xs:sequence>
1425         <xs:element name="roadOrCarriagewayOrLaneManagementType" type=
1426 ↪ "D2LogicalModel:RoadOrCarriagewayOrLaneManagementTypeEnum" minOccurs="1" maxOccurs=
1427 ↪ "1" />
1428         <xs:element name="minimumCarOccupancy" type=
1429 ↪ "D2LogicalModel:NonNegativeInteger" minOccurs="0" maxOccurs="1" />
1430         <xs:element name="roadOrCarriagewayOrLaneManagementExtension" type=
1431 ↪ "D2LogicalModel:_ExtensionType" minOccurs="0" />
1432       </xs:sequence>
1433     </xs:extension>
1434   </xs:complexContent>
1435 </xs:complexType>
1436 <xs:simpleType name="RoadOrCarriagewayOrLaneManagementTypeEnum">
1437   <xs:restriction base="xs:string">
1438     <xs:enumeration value="carPoolLaneInOperation" />
1439     <xs:enumeration value="carriagewayClosures" />
1440     <xs:enumeration value="clearALaneForEmergencyVehicles" />
1441     <xs:enumeration value="clearALaneForSnowploughsAndGrittingVehicles" />
1442     <xs:enumeration value="closedPermanentlyForTheWinter" />
1443     <xs:enumeration value="contraflow" />
1444     <xs:enumeration value="doNotUseSpecifiedLanesOrCarriageways" />
1445     <xs:enumeration value="hardShoulderRunningInOperation" />
1446     <xs:enumeration value="heightRestrictionInOperation" />
1447     <xs:enumeration value="intermittentShortTermClosures" />
1448     <xs:enumeration value="keepToTheLeft" />
1449     <xs:enumeration value="keepToTheRight" />
1450     <xs:enumeration value="laneClosures" />
1451     <xs:enumeration value="lanesDeviated" />
1452     <xs:enumeration value="narrowLanes" />
1453     <xs:enumeration value="newRoadworksLayout" />
1454     <xs:enumeration value="overnightClosures" />
1455     <xs:enumeration value="roadCleared" />
1456     <xs:enumeration value="roadClosed" />
1457     <xs:enumeration value="rollingRoadBlock" />
1458     <xs:enumeration value="rushHourLaneInOperation" />
1459     <xs:enumeration value="singleAlternateLineTraffic" />
1460     <xs:enumeration value="tidalFlowLaneInOperation" />
1461     <xs:enumeration value="turnAroundInOperation" />
1462     <xs:enumeration value="useOfSpecifiedLanesOrCarriagewaysAllowed" />

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1459     <xs:enumeration value="useSpecifiedLanesOrCarriageways" />
1460     <xs:enumeration value="vehicleStorageInOperation" />
1461     <xs:enumeration value="weightRestrictionInOperation" />
1462     <xs:enumeration value="other" />
1463   </xs:restriction>
1464 </xs:simpleType>
1465 <xs:complexType name="RoadsideAssistance">
1466   <xs:complexContent>
1467     <xs:extension base="D2LogicalModel:OperatorAction">
1468       <xs:sequence>
1469         <xs:element name="roadsideAssistanceType" type=
1470 ↪ "D2LogicalModel:RoadsideAssistanceTypeEnum" minOccurs="1" maxOccurs="1" />
1471         <xs:element name="roadsideAssistanceExtension" type="D2LogicalModel:_
1472 ↪ ExtensionType" minOccurs="0" />
1473       </xs:sequence>
1474     </xs:extension>
1475   </xs:complexContent>
1476 </xs:complexType>
1477 <xs:simpleType name="RoadsideAssistanceTypeEnum">
1478   <xs:restriction base="xs:string">
1479     <xs:enumeration value="airAmbulance" />
1480     <xs:enumeration value="busPassengerAssistance" />
1481     <xs:enumeration value="emergencyServices" />
1482     <xs:enumeration value="firstAid" />
1483     <xs:enumeration value="foodDelivery" />
1484     <xs:enumeration value="helicopterRescue" />
1485     <xs:enumeration value="vehicleRepair" />
1486     <xs:enumeration value="vehicleRecovery" />
1487     <xs:enumeration value="other" />
1488   </xs:restriction>
1489 </xs:simpleType>
1490 <xs:complexType name="Roadworks" abstract="true">
1491   <xs:complexContent>
1492     <xs:extension base="D2LogicalModel:OperatorAction">
1493       <xs:sequence>
1494         <xs:element name="roadworksDuration" type=
1495 ↪ "D2LogicalModel:RoadworksDurationEnum" minOccurs="0" maxOccurs="1" />
1496         <xs:element name="roadworksScale" type="D2LogicalModel:RoadworksScaleEnum"
1497 ↪ minOccurs="0" maxOccurs="1" />
1498         <xs:element name="underTraffic" type="D2LogicalModel:Boolean" minOccurs="0"
1499 ↪ maxOccurs="1" />
1500         <xs:element name="urgentRoadworks" type="D2LogicalModel:Boolean" minOccurs=
1501 ↪ "0" maxOccurs="1" />
1502         <xs:element name="mobility" type="D2LogicalModel:Mobility" minOccurs="0" />
1503         <xs:element name="subjects" type="D2LogicalModel:Subjects" minOccurs="0" />
1504         <xs:element name="maintenanceVehicles" type=
1505 ↪ "D2LogicalModel:MaintenanceVehicles" minOccurs="0" />
1506         <xs:element name="roadworksExtension" type="D2LogicalModel:_ExtensionType"
1507 ↪ minOccurs="0" />
1508       </xs:sequence>
1509     </xs:extension>
1510   </xs:complexContent>
1511 </xs:complexType>
1512 <xs:simpleType name="RoadworksDurationEnum">
1513   <xs:restriction base="xs:string">
1514     <xs:enumeration value="longTerm" />

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1507     <xs:enumeration value="mediumTerm" />
1508     <xs:enumeration value="shortTerm" />
1509   </xs:restriction>
1510 </xs:simpleType>
1511 <xs:simpleType name="RoadworksScaleEnum">
1512   <xs:restriction base="xs:string">
1513     <xs:enumeration value="major" />
1514     <xs:enumeration value="medium" />
1515     <xs:enumeration value="minor" />
1516   </xs:restriction>
1517 </xs:simpleType>
1518 <xs:simpleType name="Seconds">
1519   <xs:restriction base="D2LogicalModel:Float" />
1520 </xs:simpleType>
1521 <xs:complexType name="Situation">
1522   <xs:sequence>
1523     <xs:element name="situationVersionTime" type="D2LogicalModel:DateTime"
1524     ↪ minOccurs="0" maxOccurs="1" />
1525     <xs:element name="headerInformation" type="D2LogicalModel:HeaderInformation" />
1526     <xs:element name="situationRecord" type="D2LogicalModel:SituationRecord"
1527     ↪ maxOccurs="unbounded" />
1528     <xs:element name="situationExtension" type="D2LogicalModel:_ExtensionType"
1529     ↪ minOccurs="0" />
1530   </xs:sequence>
1531   <xs:attribute name="id" type="xs:string" use="required" />
1532   <xs:attribute name="version" type="xs:string" use="required" />
1533 </xs:complexType>
1534 <xs:complexType name="SituationPublication">
1535   <xs:complexContent>
1536     <xs:extension base="D2LogicalModel:PayloadPublication">
1537       <xs:sequence>
1538         <xs:element name="situation" type="D2LogicalModel:Situation" minOccurs="0"
1539         ↪ maxOccurs="unbounded" />
1540         <xs:element name="situationPublicationExtension" type="D2LogicalModel:_
1541         ↪ ExtensionType" minOccurs="0" />
1542       </xs:sequence>
1543     </xs:extension>
1544   </xs:complexContent>
1545 </xs:complexType>
1546 <xs:complexType name="SituationRecord" abstract="true">
1547   <xs:sequence>
1548     <xs:element name="situationRecordCreationTime" type="D2LogicalModel:DateTime"
1549     ↪ minOccurs="1" maxOccurs="1" />
1550     <xs:element name="situationRecordVersionTime" type="D2LogicalModel:DateTime"
1551     ↪ minOccurs="1" maxOccurs="1" />
1552     <xs:element name="probabilityOfOccurrence" type=
1553     ↪ "D2LogicalModel:ProbabilityOfOccurrenceEnum" minOccurs="1" maxOccurs="1" />
1554     <xs:element name="source" type="D2LogicalModel:Source" minOccurs="0" />
1555     <xs:element name="validity" type="D2LogicalModel:Validity" />
1556     <xs:element name="impact" type="D2LogicalModel:Impact" minOccurs="0" />
1557     <xs:element name="cause" type="D2LogicalModel:Cause" minOccurs="0" />
1558     <xs:element name="generalPublicComment" type="D2LogicalModel:Comment" minOccurs=
1559     ↪ "0" maxOccurs="unbounded" />
1560     <xs:element name="groupOfLocations" type="D2LogicalModel:GroupOfLocations" />
1561     <xs:element name="situationRecordExtension" type="D2LogicalModel:_ExtensionType
1562     ↪ " minOccurs="0" />

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1553     </xs:sequence>
1554     <xs:attribute name="id" type="xs:string" use="required" />
1555     <xs:attribute name="version" type="xs:string" use="required" />
1556   </xs:complexType>
1557   <xs:complexType name="SJTSKPointCoordinates">
1558     <xs:sequence>
1559       <xs:element name="sjtskX" type="D2LogicalModel:Float" minOccurs="1" maxOccurs="1"
1560       ↪ " />
1561       <xs:element name="sjtskY" type="D2LogicalModel:Float" minOccurs="1" maxOccurs="1"
1562       ↪ " />
1563       <xs:element name="sjtskPointCoordinatesExtension" type="D2LogicalModel:_
1564       ↪ ExtensionType" minOccurs="0" />
1565     </xs:sequence>
1566   </xs:complexType>
1567   <xs:complexType name="Source">
1568     <xs:sequence>
1569       <xs:element name="sourceCountry" type="D2LogicalModel:CountryEnum" minOccurs="0"
1570       ↪ " maxOccurs="1" />
1571       <xs:element name="sourceIdentification" type="D2LogicalModel:String" minOccurs=
1572       ↪ "0" maxOccurs="1" />
1573       <xs:element name="sourceName" type="D2LogicalModel:MultilingualString"
1574       ↪ minOccurs="0" maxOccurs="1" />
1575       <xs:element name="sourceType" type="D2LogicalModel:SourceTypeEnum" minOccurs="0"
1576       ↪ " maxOccurs="1" />
1577       <xs:element name="reliable" type="D2LogicalModel:Boolean" minOccurs="0"
1578       ↪ maxOccurs="1" />
1579       <xs:element name="sourceExtension" type="D2LogicalModel:_ExtensionType"
1580       ↪ minOccurs="0" />
1581     </xs:sequence>
1582   </xs:complexType>
1583   <xs:simpleType name="SourceTypeEnum">
1584     <xs:restriction base="xs:string">
1585       <xs:enumeration value="automobileClubPatrol" />
1586       <xs:enumeration value="cameraObservation" />
1587       <xs:enumeration value="freightVehicleOperator" />
1588       <xs:enumeration value="inductionLoopMonitoringStation" />
1589       <xs:enumeration value="infraredMonitoringStation" />
1590       <xs:enumeration value="microwaveMonitoringStation" />
1591       <xs:enumeration value="mobileTelephoneCaller" />
1592       <xs:enumeration value="nonPoliceEmergencyServicePatrol" />
1593       <xs:enumeration value="otherInformation" />
1594       <xs:enumeration value="otherOfficialVehicle" />
1595       <xs:enumeration value="policePatrol" />
1596       <xs:enumeration value="privateBreakdownService" />
1597       <xs:enumeration value="publicAndPrivateUtilities" />
1598       <xs:enumeration value="registeredMotoristObserver" />
1599       <xs:enumeration value="roadAuthorities" />
1600       <xs:enumeration value="roadOperatorPatrol" />
1601       <xs:enumeration value="roadsideTelephoneCaller" />
1602       <xs:enumeration value="spotterAircraft" />
1603       <xs:enumeration value="trafficMonitoringStation" />
1604       <xs:enumeration value="transitOperator" />
1605       <xs:enumeration value="vehicleProbeMeasurement" />
1606       <xs:enumeration value="videoProcessingMonitoringStation" />
1607     </xs:restriction>
1608   </xs:simpleType>

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1600 <xs:complexType name="SpeedManagement">
1601   <xs:complexContent>
1602     <xs:extension base="D2LogicalModel:NetworkManagement">
1603       <xs:sequence>
1604         <xs:element name="speedManagementType" type=
1605 ↪ "D2LogicalModel:SpeedManagementTypeEnum" minOccurs="0" maxOccurs="1" />
1606         <xs:element name="temporarySpeedLimit" type=
1607 ↪ "D2LogicalModel:KilometresPerHour" minOccurs="0" maxOccurs="1" />
1608         <xs:element name="speedManagementExtension" type="D2LogicalModel:_
1609 ↪ ExtensionType" minOccurs="0" />
1610       </xs:sequence>
1611     </xs:extension>
1612   </xs:complexContent>
1613 </xs:complexType>
1614 <xs:simpleType name="SpeedManagementTypeEnum">
1615   <xs:restriction base="xs:string">
1616     <xs:enumeration value="activeSpeedControlInOperation" />
1617     <xs:enumeration value="doNotSlowdownUnnecessarily" />
1618     <xs:enumeration value="observeSpeedLimit" />
1619     <xs:enumeration value="policeSpeedChecksInOperation" />
1620     <xs:enumeration value="reduceYourSpeed" />
1621     <xs:enumeration value="speedRestrictionInOperation" />
1622     <xs:enumeration value="other" />
1623   </xs:restriction>
1624 </xs:simpleType>
1625 <xs:simpleType name="String">
1626   <xs:restriction base="xs:string">
1627     <xs:maxLength value="1024" />
1628   </xs:restriction>
1629 </xs:simpleType>
1630 <xs:complexType name="Subjects">
1631   <xs:sequence>
1632     <xs:element name="subjectTypeOfWorks" type=
1633 ↪ "D2LogicalModel:SubjectTypeOfWorksEnum" minOccurs="1" maxOccurs="1" />
1634     <xs:element name="numberOfSubjects" type="D2LogicalModel:NonNegativeInteger"
1635 ↪ minOccurs="0" maxOccurs="1" />
1636     <xs:element name="subjectsExtension" type="D2LogicalModel:_ExtensionType"
1637 ↪ minOccurs="0" />
1638   </xs:sequence>
1639 </xs:complexType>
1640 <xs:simpleType name="SubjectTypeOfWorksEnum">
1641   <xs:restriction base="xs:string">
1642     <xs:enumeration value="bridge" />
1643     <xs:enumeration value="buriedCables" />
1644     <xs:enumeration value="buriedServices" />
1645     <xs:enumeration value="crashBarrier" />
1646     <xs:enumeration value="gallery" />
1647     <xs:enumeration value="gantry" />
1648     <xs:enumeration value="gasMainWork" />
1649     <xs:enumeration value="interchange" />
1650     <xs:enumeration value="junction" />
1651     <xs:enumeration value="levelCrossing" />
1652     <xs:enumeration value="lightingSystem" />
1653     <xs:enumeration value="measurementEquipment" />
1654     <xs:enumeration value="noiseProtection" />
1655     <xs:enumeration value="road" />

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1650     <xs:enumeration value="roadsideDrains" />
1651     <xs:enumeration value="roadsideEmbankment" />
1652     <xs:enumeration value="roadsideEquipment" />
1653     <xs:enumeration value="roadSigns" />
1654     <xs:enumeration value="roundabout" />
1655     <xs:enumeration value="tollGate" />
1656     <xs:enumeration value="tunnel" />
1657     <xs:enumeration value="waterMain" />
1658     <xs:enumeration value="other" />
1659 </xs:restriction>
1660 </xs:simpleType>
1661 <xs:complexType name="SupplementaryPositionalDescription">
1662   <xs:sequence>
1663     <xs:element name="locationDescriptor" type=
1664     ↪ "D2LogicalModel:LocationDescriptorEnum" minOccurs="0" maxOccurs="unbounded" />
1665     <xs:element name="sequentialRampNumber" type="D2LogicalModel:NonNegativeInteger
1666     ↪ " minOccurs="0" maxOccurs="1" />
1667     <xs:element name="affectedCarriagewayAndLanes" type=
1668     ↪ "D2LogicalModel:AffectedCarriagewayAndLanes" minOccurs="0" maxOccurs="unbounded" />
1669     <xs:element name="supplementaryPositionalDescriptionExtension" type=
1670     ↪ "D2LogicalModel:_ExtensionType" minOccurs="0" />
1671   </xs:sequence>
1672   <xs:attribute name="locationPrecision" type=
1673   ↪ "D2LogicalModel:MetresAsNonNegativeInteger" use="optional" />
1674 </xs:complexType>
1675 <xs:simpleType name="Tonnes">
1676   <xs:restriction base="D2LogicalModel:Float" />
1677 </xs:simpleType>
1678 <xs:simpleType name="TrafficConstrictionTypeEnum">
1679   <xs:restriction base="xs:string">
1680     <xs:enumeration value="carriagewayBlocked" />
1681     <xs:enumeration value="carriagewayPartiallyObstructed" />
1682     <xs:enumeration value="lanesBlocked" />
1683     <xs:enumeration value="lanesPartiallyObstructed" />
1684     <xs:enumeration value="roadBlocked" />
1685     <xs:enumeration value="roadPartiallyObstructed" />
1686   </xs:restriction>
1687 </xs:simpleType>
1688 <xs:simpleType name="TrafficStatusEnum">
1689   <xs:restriction base="xs:string">
1690     <xs:enumeration value="impossible" />
1691     <xs:enumeration value="congested" />
1692     <xs:enumeration value="heavy" />
1693     <xs:enumeration value="freeFlow" />
1694     <xs:enumeration value="unknown" />
1695   </xs:restriction>
1696 </xs:simpleType>
1697 <xs:complexType name="TrafficStatusValue">
1698   <xs:complexContent>
1699     <xs:extension base="D2LogicalModel:DataValue">
1700       <xs:sequence>
1701         <xs:element name="trafficStatusValue" type="D2LogicalModel:TrafficStatusEnum
1702         ↪ " minOccurs="1" maxOccurs="1" />
1703         <xs:element name="trafficStatusValueExtension" type="D2LogicalModel:_
1704         ↪ ExtensionType" minOccurs="0" />
1705       </xs:sequence>

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1699     </xs:extension>
1700 </xs:complexContent>
1701 </xs:complexType>
1702 <xs:simpleType name="TrafficTypeEnum">
1703   <xs:restriction base="xs:string">
1704     <xs:enumeration value="accessOnlyTraffic" />
1705     <xs:enumeration value="destinedForAirport" />
1706     <xs:enumeration value="destinedForAirportArrivals" />
1707     <xs:enumeration value="destinedForAirportDepartures" />
1708     <xs:enumeration value="destinedForFerryService" />
1709     <xs:enumeration value="destinedForRailService" />
1710     <xs:enumeration value="holidayTraffic" />
1711     <xs:enumeration value="localTraffic" />
1712     <xs:enumeration value="longDistanceTraffic" />
1713     <xs:enumeration value="regionalTraffic" />
1714     <xs:enumeration value="residentsOnlyTraffic" />
1715     <xs:enumeration value="throughTraffic" />
1716     <xs:enumeration value="visitorTraffic" />
1717   </xs:restriction>
1718 </xs:simpleType>
1719 <xs:simpleType name="UrgencyEnum">
1720   <xs:restriction base="xs:string">
1721     <xs:enumeration value="extremelyUrgent" />
1722     <xs:enumeration value="urgent" />
1723     <xs:enumeration value="normalUrgency" />
1724   </xs:restriction>
1725 </xs:simpleType>
1726 <xs:complexType name="Validity">
1727   <xs:sequence>
1728     <xs:element name="validityStatus" type="D2LogicalModel:ValidityStatusEnum"
1729 ↪ minOccurs="1" maxOccurs="1" />
1730     <xs:element name="validityTimeSpecification" type="D2LogicalModel:OverallPeriod
1731 ↪ " />
1732     <xs:element name="validityExtension" type="D2LogicalModel:_ExtensionType"
1733 ↪ minOccurs="0" />
1734   </xs:sequence>
1735 </xs:complexType>
1736 <xs:simpleType name="ValidityStatusEnum">
1737   <xs:restriction base="xs:string">
1738     <xs:enumeration value="active" />
1739     <xs:enumeration value="suspended" />
1740     <xs:enumeration value="definedByValidityTimeSpec" />
1741   </xs:restriction>
1742 </xs:simpleType>
1743 <xs:complexType name="VehicleCharacteristics">
1744   <xs:sequence>
1745     <xs:element name="loadType" type="D2LogicalModel:LoadTypeEnum" minOccurs="0"
1746 ↪ maxOccurs="1" />
1747     <xs:element name="vehicleType" type="D2LogicalModel:VehicleTypeEnum" minOccurs=
1748 ↪ "0" maxOccurs="unbounded" />
1749     <xs:element name="vehicleUsage" type="D2LogicalModel:VehicleUsageEnum"
1750 ↪ minOccurs="0" maxOccurs="1" />
1751     <xs:element name="grossWeightCharacteristic" type=
1752 ↪ "D2LogicalModel:GrossWeightCharacteristic" minOccurs="0" maxOccurs="2" />
1753     <xs:element name="heightCharacteristic" type=
1754 ↪ "D2LogicalModel:HeightCharacteristic" minOccurs="0" maxOccurs="2" />

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1747     <xs:element name="lengthCharacteristic" type=
1748 ↪ "D2LogicalModel:LengthCharacteristic" minOccurs="0" maxOccurs="2" />
1749     <xs:element name="widthCharacteristic" type="D2LogicalModel:WidthCharacteristic"
1750 ↪ minOccurs="0" maxOccurs="2" />
1751     <xs:element name="heaviestAxleWeightCharacteristic" type=
1752 ↪ "D2LogicalModel:HeaviestAxleWeightCharacteristic" minOccurs="0" maxOccurs="2" />
1753     <xs:element name="numberOfAxlesCharacteristic" type=
1754 ↪ "D2LogicalModel:NumberOfAxlesCharacteristic" minOccurs="0" maxOccurs="2" />
1755     <xs:element name="vehicleCharacteristicsExtension" type="D2LogicalModel:_
1756 ↪ ExtensionType" minOccurs="0" />
1757     </xs:sequence>
1758 </xs:complexType>
1759 <xs:complexType name="VehicleCountValue">
1760 <xs:complexContent>
1761 <xs:extension base="D2LogicalModel:DataValue">
1762 <xs:sequence>
1763 <xs:element name="vehicleCount" type="D2LogicalModel:NonNegativeInteger"
1764 ↪ minOccurs="1" maxOccurs="1" />
1765 <xs:element name="vehicleCountValueExtension" type="D2LogicalModel:_
1766 ↪ ExtensionType" minOccurs="0" />
1767 </xs:sequence>
1768 </xs:extension>
1769 </xs:complexContent>
1770 </xs:complexType>
1771 <xs:complexType name="VehicleFlowValue">
1772 <xs:complexContent>
1773 <xs:extension base="D2LogicalModel:DataValue">
1774 <xs:sequence>
1775 <xs:element name="vehicleFlowRate" type="D2LogicalModel:VehiclesPerHour"
1776 ↪ minOccurs="1" maxOccurs="1" />
1777 <xs:element name="vehicleFlowValueExtension" type="D2LogicalModel:_
1778 ↪ ExtensionType" minOccurs="0" />
1779 </xs:sequence>
1780 </xs:extension>
1781 </xs:complexContent>
1782 </xs:complexType>
1783 <xs:simpleType name="VehiclesPerHour">
1784 <xs:restriction base="D2LogicalModel:NonNegativeInteger" />
1785 </xs:simpleType>
1786 <xs:simpleType name="VehicleTypeEnum">
1787 <xs:restriction base="xs:string">
1788 <xs:enumeration value="agriculturalVehicle" />
1789 <xs:enumeration value="anyVehicle" />
1790 <xs:enumeration value="articulatedVehicle" />
1791 <xs:enumeration value="bicycle" />
1792 <xs:enumeration value="bus" />
1793 <xs:enumeration value="car" />
1794 <xs:enumeration value="caravan" />
1795 <xs:enumeration value="carOrLightVehicle" />
1796 <xs:enumeration value="carWithCaravan" />
1797 <xs:enumeration value="carWithTrailer" />
1798 <xs:enumeration value="constructionOrMaintenanceVehicle" />
1799 <xs:enumeration value="fourWheelDrive" />
1800 <xs:enumeration value="highSidedVehicle" />
1801 <xs:enumeration value="lorry" />
1802 <xs:enumeration value="moped" />

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1794     <xs:enumeration value="motorcycle" />
1795     <xs:enumeration value="motorcycleWithSideCar" />
1796     <xs:enumeration value="motorscooter" />
1797     <xs:enumeration value="tanker" />
1798     <xs:enumeration value="threeWheeledVehicle" />
1799     <xs:enumeration value="trailer" />
1800     <xs:enumeration value="tram" />
1801     <xs:enumeration value="twoWheeledVehicle" />
1802     <xs:enumeration value="van" />
1803     <xs:enumeration value="vehicleWithCatalyticConverter" />
1804     <xs:enumeration value="vehicleWithoutCatalyticConverter" />
1805     <xs:enumeration value="vehicleWithCaravan" />
1806     <xs:enumeration value="vehicleWithTrailer" />
1807     <xs:enumeration value="withEvenNumberedRegistrationPlates" />
1808     <xs:enumeration value="withOddNumberedRegistrationPlates" />
1809     <xs:enumeration value="other" />
1810   </xs:restriction>
1811 </xs:simpleType>
1812 <xs:simpleType name="VehicleUsageEnum">
1813   <xs:restriction base="xs:string">
1814     <xs:enumeration value="agricultural" />
1815     <xs:enumeration value="commercial" />
1816     <xs:enumeration value="emergencyServices" />
1817     <xs:enumeration value="military" />
1818     <xs:enumeration value="nonCommercial" />
1819     <xs:enumeration value="patrol" />
1820     <xs:enumeration value="recoveryServices" />
1821     <xs:enumeration value="roadMaintenanceOrConstruction" />
1822     <xs:enumeration value="roadOperator" />
1823     <xs:enumeration value="taxi" />
1824   </xs:restriction>
1825 </xs:simpleType>
1826 <xs:complexType name="WidthCharacteristic">
1827   <xs:sequence>
1828     <xs:element name="comparisonOperator" type=
1829 ↪ "D2LogicalModel:ComparisonOperatorEnum" minOccurs="1" maxOccurs="1" />
1830     <xs:element name="vehicleWidth" type="D2LogicalModel:MetresAsFloat" minOccurs="1"
1831 ↪ " maxOccurs="1" />
1832     <xs:element name="widthCharacteristicExtension" type="D2LogicalModel:_
1833 ↪ ExtensionType" minOccurs="0" />
1834   </xs:sequence>
1835 </xs:complexType>
1836 <xs:complexType name="WinterDrivingManagement">
1837   <xs:complexContent>
1838     <xs:extension base="D2LogicalModel:NetworkManagement">
1839       <xs:sequence>
1840         <xs:element name="winterEquipmentManagementType" type=
1841 ↪ "D2LogicalModel:WinterEquipmentManagementTypeEnum" minOccurs="1" maxOccurs="1" />
1842         <xs:element name="winterDrivingManagementExtension" type="D2LogicalModel:_
1843 ↪ ExtensionType" minOccurs="0" />
1844       </xs:sequence>
1845     </xs:extension>
1846   </xs:complexContent>
1847 </xs:complexType>
1848 <xs:simpleType name="WinterEquipmentManagementTypeEnum">
1849   <xs:restriction base="xs:string">

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```
1845 <xs:enumeration value="doNoUseStudTyres" />
1846 <xs:enumeration value="useSnowChains" />
1847 <xs:enumeration value="useSnowChainsOrTyres" />
1848 <xs:enumeration value="useSnowTyres" />
1849 <xs:enumeration value="winterEquipmentOnBoardRequired" />
1850 <xs:enumeration value="other" />
1851 </xs:restriction>
1852 </xs:simpleType>
1853 </xs:schema>
```

## CHANGELOG

version 1.1.0rc

- added feed type to schema and samples
- added OpenLR location to schema

version 1.0.1

- revision of the documentation, proofreading, changes to examples

version 1.0.0

- Initial format and documentation



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